Numerical

Prove di simulazione con soluzioni

Parte I: stato delle prove passate. Non e' stato trovato un fascicolo di appelli passati svolti per questo modulo.

Parte II: venti mock generati con testi integrali e soluzioni complete. La distinzione fra testi e soluzioni nel fascicolo e' conservata.

I vecchi appelli NSMF di inferenza statistica sono nel volume Statistica.

Parte I

Prove passate: materiale non trovato

Non sono state aggiunte prove passate non presenti nelle fonti. Per esercitarsi usare i venti mock svolti della Parte II e le pratiche negli appunti.

Parte II

20 mock generati con soluzioni

Tutti i venti testi e tutte le soluzioni del fascicolo preesistente sono conservati integralmente.

Numerical and Statistical Methods for Finance

Module in Simulation – 20 Practice Mock Exams

Complete version with full solutions and review log

Structure of this file

- Part I: complete mock exam text, including Mock Exams 1–20.
- Part II: complete worked solutions, including the solutions to Mock Exams 1–20.
- A separate review file documents three checking passes on the generated solutions.

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Part I – Mock Exam Text

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 1

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $U \sim \text{Uniform}(0,1)$.

(a) Show that $X = 1 - \sqrt{U}$ has density $f_X(x) = 2(1-x)\mathbf{1}_{[0,1]}(x)$. Identify the Beta distribution of X.
(b) Describe an inverse-transform algorithm to sample from this distribution.
(c) Let $Y_1, Y_2 \sim \text{Exp}(1)$ be independent and set $Z = Y_1/(Y_1+Y_2)$. Find the distribution of Z by using the transformation $(y_1, y_2) \mapsto (z, s) = (y_1/(y_1+y_2), y_1+y_2)$.
(d) Which of the two simulation procedures in (b) and (c) would you prefer in practice? Explain in terms of computational cost.

Question 2

Consider the integral $I = \int_0^1 \log(1+x)\,dx$.

(a) Write a vanilla Monte Carlo estimator of I, specifying target distribution and test function.
(b) Compute I exactly.
(c) Propose an antithetic estimator based on uniforms. Explain why it is expected to reduce variance.
(d) Five independent estimates of I are 0.388, 0.392, 0.384, 0.401, 0.395. Compute the jackknife estimate of the variance of their mean.

Question 3

Consider self-normalised importance sampling for estimating $\pi(\varphi)$, where the target is known only up to a normalising constant: $\gamma(x) = e^{-x}\mathbf{1}_{x\geq 0}$, and the proposal is $q(x) = 2e^{-2x}\mathbf{1}_{x\geq 0}$.

(a) Write the unnormalised weights $\widetilde{w}(x) = \gamma(x)/q(x)$.
(b) Write the SNIS estimator for $\varphi(x) = x$.
(c) The following Python code is used after sampling x from q. Explain what each line computes and why the final estimate is self-normalised.

```
logw = -x - (np.log(2) - 2*x)
a = np.max(logw)
w = np.exp(logw - a)
est = np.sum(w * x) / np.sum(w)
ess = (np.sum(w)**2) / np.sum(w**2)
```

(d) If ESS $= 95$ with $N = 1000$, what does this suggest about the proposal?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 2

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $Y \sim \text{Exp}(\lambda)$ and fix $0 < a < b$. Let X have the conditional law of Y given $a \leq Y \leq b$.

(a) Compute the cdf $F_X(x)$ for $x \in [a, b]$.
(b) Derive $F_X^{-1}(u)$ and write pseudocode to sample X from one uniform random variable.
(c) Describe a rejection sampler using $Y \sim \text{Exp}(\lambda)$ as proposal. Give the acceptance probability and the expected number of trials.
(d) For which values of a, b, λ is rejection inefficient?

Question 2

A sample of losses is $12, 15, 18, 19, 21, 44$. We estimate the typical loss using the sample median.

(a) Describe the nonparametric bootstrap algorithm for estimating the bias and standard error of the sample median.
(b) Write the formulas for the bootstrap bias and bootstrap standard error.
(c) Suppose the bootstrap medians have empirical quantiles $q_{0.025} = 15$ and $q_{0.975} = 32.5$. Give the percentile 95% confidence interval.
(d) The number of bootstrap replications used is $B = 10000$. Is this always meaningful when $n = 6$? Explain by referring to the number of distinct bootstrap samples.

Question 3

Let $X \sim \pi$, where $\pi(x) = \frac{1}{2}\mathbf{1}_{[-1,1]}(x)$, and consider $\theta = \mathbb{E}_\pi[e^X]$.

(a) Write a vanilla Monte Carlo estimator of θ and prove that it is unbiased.
(b) Propose a control variate based on X. State the optimal coefficient.
(c) Compute $\mathbb{E}_\pi[X]$ and $\text{Var}_\pi(X)$.
(d) Explain why the control variate is useful for e^X.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 3

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Consider a discrete random variable $X \in \{0, 1, 2, 3\}$ with probabilities $p_0 = 0.1$, $p_1 = 0.2$, $p_2 = 0.4$, $p_3 = 0.3$.

(a) Describe the inverse transform method for a general finite discrete distribution.
(b) Apply the method to $u = 0.63$.
(c) Write pseudocode for the discrete inverse transform method.
(d) Explain why the naive algorithm becomes inefficient when the number of states is large.

Question 2

Let X have unnormalised density $\gamma(x) = x^2(1-x)^2\mathbf{1}_{[0,1]}(x)$.

(a) Find the normalising constant and identify the corresponding Beta distribution.
(b) Construct a rejection sampler using $q(x) = \mathbf{1}_{[0,1]}(x)$. Find the smallest valid M.
(c) Compute the acceptance rate.
(d) Write pseudocode for the rejection sampler.

Question 3

Consider the following Python code.

```
u = np.random.rand(N)
x = np.tan(np.pi * (u - 0.5))
y = x[x > 3]
p_hat = len(y) / N
```

(a) Which distribution is simulated by the transformation defining x?
(b) What probability is *p_hat* estimating?
(c) Give the exact value of the probability in terms of the cdf of the simulated distribution.
(d) Suggest one variance reduction idea for estimating this tail probability, without writing Python code.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 4

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $X \sim \mathcal{N}(0,1)$ and let $Y \mid X = x \sim \mathcal{N}(x,1)$.

(a) Write the joint density $\pi(x,y)$.
(b) Describe an algorithm to sample from the joint distribution.
(c) Find the marginal distribution of Y.
(d) Find the conditional distribution of $X \mid Y = y$ by completing the square.

Question 2

Let $I = \int_0^{+\infty} e^{-x} \cos(x)\, dx$.

(a) Express I as $\mathbb{E}[\varphi(X)]$ for a suitable distribution of X.
(b) Write a Monte Carlo estimator and its variance.
(c) Compute I exactly.
(d) Explain why sampling directly on $[0,+\infty)$ using the exponential density is better than truncating the integral at a large constant without justification.

Question 3

Suppose we want to estimate $\mathbb{P}(Z > 4)$, where $Z \sim \mathcal{N}(0,1)$, using importance sampling with proposal $q = \mathcal{N}(\mu,1)$.

(a) Write the IS estimator.
(b) Derive the weight $w(x) = \pi(x)/q(x)$.
(c) Explain why choosing $\mu > 0$ may reduce variance.
(d) What may go wrong if μ is chosen too large?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 5

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $U_1, U_2 \sim \text{Uniform}(0,1)$ independent and define $R = \sqrt{U_1}$, $\Theta = 2\pi U_2$, $X = R\cos\Theta$, $Y = R\sin\Theta$.

(a) Show that (X, Y) is uniformly distributed on the unit disk.
(b) Explain the role of the transformation $R = \sqrt{U_1}$ instead of $R = U_1$.
(c) Write pseudocode to generate N independent points uniformly on the disk.
(d) How would the algorithm change for a disk of radius r centred at (a, b)?

Question 2

A statistic $T = T(X_1, \ldots, X_n)$ is estimated by jackknife.

(a) Define the leave-one-out estimates $T_{(-i)}$ and the jackknife average $\overline{T}_{(\cdot)}$.
(b) Write the jackknife bias estimate and standard error estimate.
(c) For the observations $2, 4, 5, 9$, compute the jackknife leave-one-out estimates of the sample mean.
(d) Compute the jackknife estimate of the variance of the sample mean from these leave-one-out estimates.

Question 3

Let $\pi(x) \propto e^{-x^2/2}$ and use the double-exponential proposal $q_\lambda(x) = \frac{\lambda}{2} e^{-\lambda|x|}$.

(a) Write the ratio $\gamma(x)/q_\lambda(x)$, where $\gamma(x) = e^{-x^2/2}$.
(b) Find the value of $x \geq 0$ where this ratio is maximised.
(c) Find the smallest valid rejection constant M_λ.
(d) Explain how one would choose λ to improve the rejection sampler.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 6

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $X \mid P = p \sim \text{Binomial}(m, p)$, where $P \sim \text{Beta}(\alpha, \beta)$.

(a) Describe a composition algorithm to sample X.
(b) Write the marginal probability $\mathbb{P}(X = k)$ as an integral.
(c) Explain why this is a discrete mixture.
(d) In the special case $m = 1$, compute $\mathbb{P}(X = 1)$.

Question 2

Consider estimating $\theta = \mathbb{E}[\sqrt{U}]$, where $U \sim \text{Uniform}(0, 1)$.

(a) Compute θ exactly.
(b) Write the vanilla Monte Carlo estimator.
(c) Propose a stratified sampling estimator using L equal strata on $[0, 1]$.
(d) Explain why stratifying uniforms can reduce variance for monotone functions.

Question 3

Consider the following Python output from a bootstrap procedure for the sample median:

```
B = 2000
np.quantile(theta_star, [0.025, 0.5, 0.975])
# array([4.0, 6.5, 11.0])
theta_hat = 6.5
np.mean(theta_star) - theta_hat
# 0.35
np.std(theta_star, ddof=1)
# 1.92
```

(a) What is the bootstrap estimate of the bias?
(b) What is the bootstrap estimate of the standard error?
(c) Give the percentile 95% confidence interval.
(d) Does the code construct a parametric or nonparametric bootstrap? What information is missing to answer with certainty?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 7

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let X have density $f(x) = \frac{3}{8}(x+1)^2 \mathbf{1}_{[-1,1]}(x)$.

(a) Compute the cdf $F(x)$ on $[-1, 1]$.
(b) Derive an inverse-transform method to sample X.
(c) Write pseudocode for the algorithm.
(d) Check that the density integrates to one.

Question 2

Let $I = \int_0^1 \frac{1}{1+x^2}\, dx$.

(a) Write a Monte Carlo estimator of $4I$ and identify the classical constant being estimated.
(b) Compute the exact variance of the estimator based on N iid uniforms.
(c) Propose an antithetic estimator and write it explicitly.
(d) Explain how to estimate the standard error from simulation output.

Question 3

For importance sampling, suppose $X_i \sim q$, $w(x) = \pi(x)/q(x)$, and $\widehat{\theta}_N = N^{-1} \sum_{i=1}^N w(X_i)\varphi(X_i)$.

(a) State the support condition required for valid importance sampling.
(b) Prove that $\widehat{\theta}_N$ is unbiased.
(c) Show that $\mathbb{E}_q[w(X)] = 1$.
(d) Explain the difference between IS and SNIS when the normalising constant of π is unknown.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 8

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $X \sim \mathcal{N}(0,1)$. We want to sample from $X \mid X > c$, with $c > 0$.

(a) Write the target density.
(b) Describe a naive rejection sampler using $\mathcal{N}(0,1)$ as proposal and compute its acceptance probability.
(c) Explain why the sampler is inefficient for large c.
(d) Suggest a better proposal and state what condition the rejection constant must satisfy.

Question 2

Let $X_1, \ldots, X_n$ be iid with unknown distribution. Let $\widehat{F}_n$ be the empirical cdf.

(a) Define $\widehat{F}_n(x)$.
(b) Explain the plug-in principle.
(c) Use the plug-in principle to estimate $\mathbb{P}(X \leq t)$, $\mathbb{E}[X]$, and $\mathrm{Var}(X)$.
(d) Explain how the empirical quantile is obtained from $\widehat{F}_n$.

Question 3

Consider the following algorithm.

```
For i = 1,...,N:
   Draw X_i from q
   Draw U_i from Uniform(0,1)
   If U_i <= gamma(X_i)/(M q(X_i)):
      accept X_i
   otherwise:
      reject X_i
```

(a) Which simulation method is described?
(b) State the condition on M.
(c) Prove that the accepted samples have density proportional to γ.
(d) What is the acceptance rate when γ has normalising constant z?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 9

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

A random variable X has logistic cdf $F(x) = 1/(1 + e^{-x})$, $x \in \mathbb{R}$.

(a) Derive the inverse cdf.
(b) Write pseudocode to generate X from one uniform random variable.
(c) Compute the density by differentiating the cdf.
(d) Explain why inverse transform is convenient in this case.

Question 2

Let $Y \sim \text{Poisson}(\Lambda)$, where $\Lambda \sim \text{Gamma}(\alpha, \beta)$, with rate parameter β.

(a) Describe a composition algorithm to simulate Y.
(b) Write $\mathbb{P}(Y = k)$ as an integral over λ.
(c) Identify the type of mixture.
(d) Explain in words why this model can produce overdispersion relative to a Poisson with fixed parameter.

Question 3

Suppose we estimate $\mathbb{E}_\pi[X^2]$ using SNIS. The unnormalised weights from $N = 8$ simulations are $10, 1, 1, 1, 1, 1, 1, 1$.

(a) Compute the ESS.
(b) Interpret the result.
(c) Explain the relation between weight degeneracy and variance.
(d) Suggest one way to improve the proposal.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 10

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let (X, Y) have joint density $f(x, y) = 2\mathbf{1}_{0<y<x<1}$.

(a) Find the marginal density of X.
(b) Find the conditional density of $Y \mid X = x$.
(c) Describe an algorithm to sample from the joint distribution using conditional simulation.
(d) Check that the joint density integrates to one.

Question 2

Consider $\theta = \mathbb{P}(X > 2)$, where $X \sim \text{Exp}(1)$.

(a) Estimate θ by vanilla Monte Carlo and give the variance of the estimator.
(b) Use importance sampling with proposal $q_\lambda(x) = \lambda e^{-\lambda x}\mathbf{1}_{x\geq 0}$. Write the estimator.
(c) Derive the weight $w_\lambda(x)$.
(d) Explain qualitatively how λ should be chosen for a right-tail probability.

Question 3

Consider a variational approximation problem. The target posterior is $p(z \mid y) = \mathcal{N}(y/2, 1/2)$, and the approximating family is $q_{a,b}(z) = \mathcal{N}(ay + b, 2/3)$.

(a) Write the formula for $KL(q_{a,b} \| p)$ between two univariate Gaussian densities.
(b) Identify which term depends on a, b.
(c) Find the values of a, b that minimise the KL divergence for all y.
(d) Explain what would change if the variance of q were also a free parameter.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 11

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $y_1, \ldots, y_n$ be iid observations from an unknown distribution F, and let $\widehat{F}_n$ be the empirical cdf.

(a) Define $\widehat{F}_n(y)$ and state two of its basic properties.
(b) For the data $3, 5, 5, 8, 11$, compute $\widehat{F}_n(5)$, $\widehat{F}_n(7)$, and $\widehat{F}_n(11)$.
(c) Use the plug-in principle to estimate $\theta = \mathbb{P}(Y \leq 6)$, $\mu = \mathbb{E}[Y]$, and $\sigma^2 = \mathrm{Var}(Y)$.
(d) Explain how a histogram estimator differs from the empirical cdf and why the bin width creates a bias–variance tradeoff.

Question 2

Consider the linear congruential generator $x_{k+1} = (ax_k + c) \bmod m$, $u_k = x_k/m$, with $m = 16$, $a = 5$, $c = 1$, and $x_0 = 3$.

(a) Compute x_1, x_2, x_3, x_4 and the corresponding uniforms.
(b) What does it mean for a pseudo-random sequence to have a period?
(c) The following Python code is used to generate points. Explain what the code does and what defect may appear in low-quality LCGs.

```
x = 1
u = []
for k in range(3000):
    x = (65539 * x) % (2**31)
    u.append(x / 2**31)
points = [(u[i], u[i+1], u[i+2]) for i in range(0, len(u)-2, 3)]
```

(d) Why is visual inspection of pairs or triples of uniforms useful but not sufficient to prove that a generator is good?

Question 3

Let $\widehat{\theta}$ be a statistic and suppose that B bootstrap replications $\widehat{\theta}^*_1, \ldots, \widehat{\theta}^*_B$ are available.

(a) Define the bootstrap estimate of bias and standard error.
(b) Define the percentile bootstrap confidence interval with nominal level 95%.
(c) Suppose $\widehat{\theta} = 10$, $\overline{\theta^*} = 10.4$, $s_{\theta^*} = 1.8$, and the empirical 2.5% and 97.5% quantiles are 7.2 and 13.9. Give the bias estimate, standard error estimate, and percentile interval.
(d) Explain the difference between nonparametric and parametric bootstrap in this setting.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 12

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $U \sim \text{Uniform}(0,1)$ and let $X = \alpha U^{1/\beta}$, where $\alpha > 0$, $\beta > 0$.

(a) Find the cdf and density of X.
(b) Identify a simple inverse-transform algorithm to sample a random variable with density proportional to $x^{\beta-1}\mathbf{1}_{[0,\alpha]}(x)$.
(c) For $\alpha = 2$ and $\beta = 3$, compute $\mathbb{P}(X \leq 1)$.
(d) Explain why the inverse-transform method is exact in principle but may still be inconvenient for some distributions.

Question 2

Let $Y_1, \ldots, Y_\alpha \sim \text{Exp}(1)$ and $Z_1, \ldots, Z_\beta \sim \text{Exp}(1)$, all independent, where α, β are positive integers. Define $S = \sum_{i=1}^{\alpha} Y_i$, $T = \sum_{j=1}^{\beta} Z_j$, and $R = S/(S+T)$.

(a) State the distributions of S and T.
(b) Show, using a suitable transformation, that R follows a Beta distribution. You may leave the normalising constant in Gamma-function notation.
(c) Write pseudocode to sample $\text{Beta}(\alpha, \beta)$ using exponential random variables.
(d) Compare this method with direct inverse transform when $\alpha = \beta = 1$.

Question 3

Consider $I = \int_0^1 x^2 e^{-x}\, dx$.

(a) Write I as an expectation under a uniform distribution.
(b) Write a vanilla Monte Carlo estimator of I.
(c) Propose a control variate based on X, where $X \sim \text{Uniform}(0,1)$, and state the optimal control coefficient.
(d) The following Python code is used after a simulation. Explain what it computes.

```
phi = x**2 * np.exp(-x)
h = x - 0.5
beta = np.cov(phi, h, ddof=1)[0,1] / np.var(h, ddof=1)
est_cv = np.mean(phi - beta*h)
```

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 13

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let the target density be $\pi(x) = 6x(1-x)\mathbf{1}_{[0,1]}(x)$.

(a) Identify π as a Beta distribution.
(b) Construct a rejection sampler with proposal $q(x) = \mathbf{1}_{[0,1]}(x)$. Find the optimal constant M.
(c) Compute the acceptance rate.
(d) Write pseudocode for the rejection sampler.

Question 2

We want to sample from the standard normal density using the Laplace proposal $q(x) = \frac{1}{2}e^{-|x|}$.

(a) Write the rejection condition in terms of $\gamma(x) = e^{-x^2/2}$ and $q(x)$.
(b) Find where $\gamma(x)/q(x)$ is maximised.
(c) Find a valid rejection constant M for the unnormalised target γ.
(d) Explain why the reverse proposal, normal proposal for a Laplace target, is problematic.

Question 3

Consider the following Python code for estimating a probability.

```
N = 10000
x = np.random.normal(loc=3.0, scale=1.0, size=N)
logw = -0.5*x**2 + 0.5*(x-3.0)**2
ind = (x > 5)
est = np.mean(np.exp(logw) * ind)
se = np.std(np.exp(logw) * ind, ddof=1) / np.sqrt(N)
```

(a) What probability is being estimated?
(b) Identify the target and proposal densities.
(c) Derive the importance weight used in the code, up to cancellation of normalising constants.
(d) Explain why shifting the normal proposal to the right can reduce variance for this problem.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 14

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let (X, Y) have joint density $f(x, y) = 3x\mathbf{1}_{0<y<x<1}$.

(a) Check whether f is normalised. If not, find the correct normalising constant.
(b) Find the marginal density of X.
(c) Find the conditional density of $Y \mid X = x$.
(d) Describe a conditional simulation algorithm for (X, Y).

Question 2

Consider a Gaussian mixture with density $\pi(x) = 0.3\mathcal{N}(x; -2, 1) + 0.7\mathcal{N}(x; 1, 4)$.

(a) Describe a composition algorithm to sample from π.
(b) Compute $\mathbb{E}[X]$.
(c) Write $\mathrm{Var}(X)$ using the law of total variance.
(d) Explain why sampling from a mixture is not the same as sampling from the weighted average of two already generated observations.

Question 3

Let $\gamma(x) = e^{-x^2/2}$ and let $q(x) = \frac{1}{\pi(1+x^2)}$ be the Cauchy density.

(a) Write an importance sampling estimator of the normalising constant $Z = \int_{-\infty}^{+\infty} \gamma(x)\, dx$.
(b) Write the corresponding weights.
(c) Is the estimator unbiased? Explain.
(d) Would this proposal have heavier or lighter tails than the target? Why does this matter?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 15

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let X have Pareto density $f(x) = \alpha x_m^\alpha x^{-\alpha-1}\mathbf{1}_{x\geq x_m}$, with $\alpha > 0$.

(a) Derive the cdf of X.
(b) Derive the inverse cdf and write pseudocode to sample from this distribution.
(c) Compute $\mathbb{P}(X > K)$ for $K \geq x_m$.
(d) Explain why Pareto samples can create high Monte Carlo variability when estimating means for small α.

Question 2

For the data $4, 7, 9, 10, 20$, let $\widehat{\theta}$ be the sample variance computed with denominator n.

(a) Compute $\widehat{\theta}$.
(b) Define the jackknife replications $\widehat{\theta}_{(-i)}$.
(c) Write the jackknife estimate of bias.
(d) Explain why the jackknife is useful for detecting finite-sample bias of nonlinear estimators.

Question 3

Let $\theta = \mathbb{E}[e^{-X}]$, where $X \sim \text{Exp}(1)$.

(a) Compute θ exactly.
(b) Write a vanilla Monte Carlo estimator and its variance.
(c) Propose an antithetic estimator using $X = F^{-1}(U)$ and $X' = F^{-1}(1-U)$.
(d) Explain why antithetic variables are useful when the transformed integrand is monotone.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 16

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $X \sim \text{Poisson}(\lambda)$.

(a) Describe the discrete inverse transform method to sample from X.
(b) For $\lambda = 2$, compute the cumulative probabilities up to $k = 3$.
(c) If $u = 0.80$, determine the sampled value using the cumulative probabilities from part (b).
(d) The following Python code is used. Explain the role of the while loop.

```
u = np.random.rand()
p = np.exp(-lam)
F = p
k = 0
while u > F:
    k = k + 1
    p = p * lam / k
    F = F + p
```

Question 2

Let $I = \int_0^1 \sqrt{x}\, dx$.

(a) Compute I exactly.
(b) Write the vanilla Monte Carlo estimator.
(c) Construct a stratified estimator with two strata $[0, 1/2]$ and $(1/2, 1]$, using one draw per stratum.
(d) Explain why stratification is expected to improve estimation for $\sqrt{x}$.

Question 3

Let $S_T = S_0 \exp((r - \sigma^2/2)T + \sigma\sqrt{T}Z)$, where $Z \sim \mathcal{N}(0,1)$. Consider estimating the discounted call price $e^{-rT}\mathbb{E}[(S_T - K)^+]$.

(a) Write the vanilla Monte Carlo estimator.
(b) Propose an antithetic estimator using Z and $-Z$.
(c) Suggest a control variate based on S_T. State its known expectation.
(d) Explain why variance reduction is important for out-of-the-money options.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 17

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let X have density $\pi(x) = \frac{1}{Z} e^{-x} \mathbf{1}_{0<x<2}$.

(a) Compute the normalising constant Z.
(b) Derive the cdf and inverse cdf.
(c) Write pseudocode to sample X by inversion.
(d) Construct a rejection sampler using $\mathrm{Exp}(1)$ as proposal and compute its acceptance rate.

Question 2

Consider estimating the median from the data $8, 9, 10, 13, 14, 60$.

(a) Explain why the median may be preferred to the mean for these data.
(b) Describe the nonparametric bootstrap procedure for the sample median.
(c) Suppose the bootstrap output is:

```
B = 4000
np.quantile(med_star, [0.025, 0.975])
# array([8.5, 37.0])
np.mean(med_star) - np.median(y)
# 3.1
np.std(med_star, ddof=1)
# 10.4
```

Give the percentile interval, bootstrap bias estimate, and bootstrap standard error.
(d) Is $B = 4000$ necessarily informative when $n = 6$? Discuss the number of distinct bootstrap samples.

Question 3

Suppose $X_i \sim q$ and $\widetilde{w}_i = \gamma(X_i)/q(X_i)$, where γ is unnormalised.

(a) Write the SNIS estimator of $\pi(\varphi)$.
(b) Explain why SNIS can be used without knowing the normalising constant of γ.
(c) State whether SNIS is unbiased for finite N.
(d) Define ESS and explain its interpretation.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 18

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Consider the two-dimensional integral $I = \int_0^1 \int_0^1 e^{xy}\, dx\, dy$.

(a) Write I as an expectation using iid uniforms.
(b) Write a Monte Carlo estimator and show that it is unbiased.
(c) Give an estimator of its Monte Carlo standard error from simulated values.
(d) Explain why the convergence rate of Monte Carlo does not directly depend on the dimension in the same way as grid quadrature.

Question 2

Let $\theta = \mathbb{E}_\pi[\varphi(X)]$, and suppose a control variate $h(X)$ satisfies $\mathbb{E}_\pi[h(X)] = 0$.

(a) Write the control variate estimator with coefficient β.
(b) Derive the optimal β^*.
(c) Show the corresponding minimum variance formula.
(d) Explain why a control variate with weak correlation may be useless.

Question 3

Let π be a target density and q_ϕ a parametric proposal family.

(a) State the ideal importance sampling proposal for estimating $\pi(\varphi)$ when $\varphi \geq 0$.
(b) Explain why the ideal proposal is usually not directly usable.
(c) Define the forward KL criterion $KL(\pi \| q_\phi)$ and the reverse KL criterion $KL(q_\phi \| \pi)$.
(d) In words, compare mass-covering and mode-seeking behaviour.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 19

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $X \sim \mathcal{N}(0,1)$ and define $Y = X^2$.

(a) Derive the density of Y by transformation.
(b) Identify the distribution of Y.
(c) Explain how this result can be used to simulate a χ^2_n random variable.
(d) Write pseudocode to simulate χ^2_n.

Question 2

Let $X \mid \Lambda = \lambda \sim \text{Poisson}(\lambda)$, and $\Lambda \sim \text{Gamma}(\alpha, \beta)$ with rate β.

(a) Write a composition algorithm to sample X.
(b) Write the marginal probability $\mathbb{P}(X = k)$ as an integral.
(c) Compute $\mathbb{E}[X]$ using iterated expectation.
(d) Compute $\text{Var}(X)$ using the law of total variance.

Question 3

We want to estimate $\mathbb{P}(X > K)$, where $X \sim \text{Exp}(1)$, using the proposal $q_\lambda(x) = \lambda e^{-\lambda x}\mathbf{1}_{x \geq 0}$.

(a) Write the importance sampling estimator.
(b) Derive the weight $w_\lambda(x)$.
(c) For a right-tail probability, should λ be smaller or larger than 1? Explain.
(d) The following code is used. Explain what changes if λ is too small.

```
x = np.random.exponential(scale=1/lam, size=N)
w = np.exp(-x) / (lam*np.exp(-lam*x))
est = np.mean(w * (x > K))
```

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

Module in Simulation — Practice Mock Exam 20

Name and Student ID: _____

Instructions. Each practice exam has 3 questions and is designed for a 1:30 hour closed-book exam. Use black or blue ink in the final copy. Calculators are allowed. Write clearly and justify every step. Python code, when present, is only to be interpreted or analysed: you are not asked to write Python code. If an algorithm is requested, write pseudocode.

Question 1

Let $U \sim \text{Uniform}(0,1)$ and define $X = \sqrt{U}$. Also let $Y_1, Y_2, Y_3 \sim \text{Exp}(1)$ independent, $T = Y_1 + Y_2$, and $Z = T/(T+Y_3)$.

(a) Show that $X \sim \text{Beta}(2,1)$.
(b) Show that T has density $te^{-t}\mathbf{1}_{t\geq 0}$.
(c) Show that $Z \sim \text{Beta}(2,1)$ using a change of variables.
(d) Which method is preferable computationally for sampling $\text{Beta}(2,1)$, and why?

Question 2

Let $I = 4\int_0^1 \sqrt{1-x^2}\,dx$.

(a) Write a Monte Carlo estimator of I and identify the target distribution and test function.
(b) Knowing that $I = \pi$, write the variance of the estimator in terms of $\mathbb{E}[16(1-X^2)] - \pi^2$, where $X \sim \text{Uniform}(0,1)$, and simplify it as much as possible.
(c) Propose an antithetic estimator.
(d) Five independent estimates are $3.05, 3.18, 3.09, 3.22, 3.11$. Write the jackknife procedure to estimate the variance of their average.

Question 3

Consider importance sampling and self-normalised importance sampling.

(a) Write the standard IS estimator of $\int \varphi(x)\pi(x)\,dx$ with proposal q.
(b) Prove that the standard IS estimator is unbiased when π is normalised and the support condition holds.
(c) Write the SNIS estimator when only $\gamma(x) = Z\pi(x)$ is known.
(d) Explain the main differences between IS and SNIS: normalising constant, bias, and use of ESS.

Part II – Mock Exam Solutions

Complete Solution – Mock Exam 1

Question 1.

(a) Since $U \in [0,1]$, also $X = 1 - \sqrt{U} \in [0,1]$. For $0 \le x \le 1$,

$$F_X(x) = \mathbb{P}(1 - \sqrt{U} \le x) = \mathbb{P}(\sqrt{U} \ge 1 - x) = \mathbb{P}(U \ge (1-x)^2).$$

Because U is uniform,

$$F_X(x) = 1 - (1-x)^2 = 2x - x^2.$$

Thus

$$f_X(x) = F_X'(x) = 2 - 2x = 2(1-x), \qquad 0 < x < 1,$$

and $f_X(x) = 2(1-x)\mathbf{1}_{[0,1]}(x)$. Comparing with $f_{\text{Beta}(\alpha,\beta)}(x) = x^{\alpha-1}(1-x)^{\beta-1}/B(\alpha,\beta)$ gives $X \sim \text{Beta}(1,2)$.

(b) The cdf is $F(x) = 2x - x^2$. To invert it, solve

$$u = 2x - x^2 \quad \Longleftrightarrow \quad x^2 - 2x + u = 0.$$

The two algebraic roots are $x = 1 \pm \sqrt{1-u}$. The only root in $[0,1]$ is

$$F^{-1}(u) = 1 - \sqrt{1-u}.$$

Algorithm: draw $V \sim U(0,1)$ and return $X = 1 - \sqrt{1-V}$. Since $1 - V \sim U(0,1)$, this is equivalent in distribution to $X = 1 - \sqrt{V}$.

(c) Define

$$z = \frac{y_1}{y_1 + y_2}, \qquad s = y_1 + y_2.$$

The inverse transformation is

$$y_1 = zs, \qquad y_2 = (1-z)s,$$

with $0 < z < 1$ and $s > 0$. The Jacobian matrix is

$$\begin{pmatrix} \partial y_1/\partial z & \partial y_1/\partial s \\ \partial y_2/\partial z & \partial y_2/\partial s \end{pmatrix} = \begin{pmatrix} s & z \\ -s & 1-z \end{pmatrix},$$

whose determinant is $s(1-z) + sz = s$. Since Y_1, Y_2 are independent exponentials, $f_{Y_1,Y_2}(y_1,y_2) = e^{-y_1}e^{-y_2} = e^{-s}$. Hence

$$f_{Z,S}(z,s) = se^{-s}\mathbf{1}_{(0,1)}(z)\mathbf{1}_{(0,\infty)}(s).$$

Integrating out s gives

$$f_Z(z) = \int_0^\infty se^{-s}\,ds = \Gamma(2) = 1, \qquad 0 < z < 1.$$

Therefore $Z \sim U(0,1) = \text{Beta}(1,1)$.

(d) The procedure in (b) simulates $\text{Beta}(1,2)$, whereas the ratio in (c) simulates $\text{Beta}(1,1)$; therefore they are not interchangeable for the same target. For $\text{Beta}(1,2)$, inversion is preferable because it requires only one uniform random variable and one square root. The exponential-ratio construction requires generating exponentials and performing a ratio, so it is computationally heavier and, in this specific case, targets a different law.

Question 2.

(a) Let $U_1, \ldots, U_N \sim U(0,1)$ iid. With target distribution $U(0,1)$ and test function $\varphi(u) = \log(1+u)$,

$$I = \int_0^1 \log(1+x)\,dx = \mathbb{E}[\varphi(U)].$$

The vanilla Monte Carlo estimator is

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^N \log(1+U_i).$$

It is unbiased because $\mathbb{E}[\widehat{I}_N] = I$.

(b) Use integration by parts or the primitive

$$\int \log(1+x)\,dx = (1+x)\log(1+x) - (1+x) + C.$$

Therefore

$$I = \big[(1+x)\log(1+x) - (1+x)\big]_0^1 = (2\log 2 - 2) - (-1) = 2\log 2 - 1.$$

(c) For m antithetic pairs, draw $U_1, \dots, U_m \sim U(0,1)$ and use the paired uniforms U_i and $1 - U_i$. The estimator is

$$\widehat{I}_A = \frac{1}{m} \sum_{i=1}^{m} \frac{\log(1+U_i) + \log(1+(1-U_i))}{2} = \frac{1}{m} \sum_{i=1}^{m} \frac{\log(1+U_i) + \log(2-U_i)}{2}.$$

The function $u \mapsto \log(1+u)$ is increasing, while $u \mapsto \log(2-u)$ is decreasing. Hence the two paired evaluations tend to move in opposite directions, producing negative covariance and reducing variance.

(d) The five estimates are $\theta_1 = 0.388$, $\theta_2 = 0.392$, $\theta_3 = 0.384$, $\theta_4 = 0.401$, $\theta_5 = 0.395$. Their mean is

$$\bar{\theta} = \frac{0.388 + 0.392 + 0.384 + 0.401 + 0.395}{5} = 0.392.$$

The sample variance of the five estimates is

$$s^2 = \frac{1}{4} \sum_{i=1}^{5} (\theta_i - \bar{\theta})^2 = \frac{0.000170}{4} = 0.0000425.$$

For the mean, the jackknife variance estimate equals the usual variance estimate of the average:

$$\widehat{\mathrm{Var}}_{jack}(\bar{\theta}) = \frac{s^2}{5} = \frac{0.0000425}{5} = 0.0000085.$$

Thus the estimated variance is $8.5 \cdot 10^{-6}$.

Question 3.

(a) For $x \geq 0$,

$$\widetilde{w}(x) = \frac{\gamma(x)}{q(x)} = \frac{e^{-x}}{2e^{-2x}} = \frac{1}{2} e^{x}.$$

For $x < 0$ both densities are outside the relevant support, so the weight is only needed on $[0, \infty)$.

(b) If $X_1, \dots, X_N \sim q$ iid, the self-normalised importance sampling estimator is

$$\widehat{\theta}_{SNIS} = \frac{\sum_{i=1}^{N} \widetilde{w}(X_i) \varphi(X_i)}{\sum_{i=1}^{N} \widetilde{w}(X_i)}.$$

For $\varphi(x) = x$ this becomes

$$\widehat{\theta}_{SNIS} = \frac{\sum_{i=1}^{N} \frac{1}{2} e^{X_i} X_i}{\sum_{i=1}^{N} \frac{1}{2} e^{X_i}} = \frac{\sum_{i=1}^{N} e^{X_i} X_i}{\sum_{i=1}^{N} e^{X_i}}.$$

The constant $1/2$ cancels because the estimator normalises by the sum of the weights.

(c) The first line computes

$$\log \widetilde{w}(x) = \log \gamma(x) - \log q(x) = -x - [\log 2 - 2x] = x - \log 2.$$

The line $a = \max(\log w)$ stores the largest log-weight. Subtracting a before exponentiating gives

$$w_i = \exp(\log \widetilde{w}_i - a),$$

which is numerically stable because the largest transformed weight is 1. The estimator

$$\texttt{est} = \frac{\sum_i w_i x_i}{\sum_i w_i}$$

is self-normalised. Multiplying every original weight by the same positive constant e^{-a} does not change this ratio. Finally,

$$\mathrm{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}$$

measures how many equally weighted samples would give roughly the same weight concentration.

(d) With $N = 1000$ but ESS $= 95$, only about 9.5% of the nominal sample size is effectively contributing. This indicates severe weight variability: the proposal $q(x) = 2e^{-2x}$ decays faster than the target proportional to e^{-x}, so it undersamples the right tail where weights $\frac{1}{2} e^{x}$ can be large.

Complete Solution – Mock Exam 2

Question 1.

(a) Let

$$D = \mathbb{P}(a \le Y \le b) = F_Y(b) - F_Y(a) = \left(1 - e^{-\lambda b}\right) - \left(1 - e^{-\lambda a}\right) = e^{-\lambda a} - e^{-\lambda b}.$$

For $x \in [a, b]$,

$$F_X(x) = \mathbb{P}(Y \le x \mid a \le Y \le b) = \frac{\mathbb{P}(a \le Y \le x)}{\mathbb{P}(a \le Y \le b)} = \frac{e^{-\lambda a} - e^{-\lambda x}}{e^{-\lambda a} - e^{-\lambda b}}.$$

Also $F_X(x) = 0$ for $x < a$ and $F_X(x) = 1$ for $x > b$.

(b) Set $u = F_X(x)$ and solve:

$$u = \frac{e^{-\lambda a} - e^{-\lambda x}}{D} \quad \Longrightarrow \quad e^{-\lambda x} = e^{-\lambda a} - uD.$$

Therefore

$$F_X^{-1}(u) = -\frac{1}{\lambda} \log\{e^{-\lambda a} - u(e^{-\lambda a} - e^{-\lambda b})\}, \qquad 0 < u < 1.$$

Pseudocode:

```
Draw U from Uniform(0,1)
D = exp(-lambda*a) - exp(-lambda*b)
X = -(1/lambda)*log(exp(-lambda*a) - U*D)
Return X
```

(c) Use the original exponential distribution as proposal and accept only if the draw falls in $[a, b]$:

```
Repeat:
    Draw Y from Exp(lambda)
Until a <= Y <= b
Return Y
```

The accepted value has exactly the conditional law of Y given $a \le Y \le b$. The acceptance probability is

$$p_{acc} = \mathbb{P}(a \le Y \le b) = e^{-\lambda a} - e^{-\lambda b} = D.$$

The number of trials is geometric with mean $1/p_{acc}$, so the expected number of proposals is $1/D$.

(d) Rejection is inefficient whenever $D = e^{-\lambda a} - e^{-\lambda b}$ is small. This happens if the interval $[a, b]$ is very narrow, or if it lies far in the right tail, or if λ and a, b make the exponential mass in $[a, b]$ tiny. In those cases many proposed exponentials are rejected before one is accepted.

Question 2.

(a) The observed sample is $y = (12, 15, 18, 19, 21, 44)$ and the statistic is the sample median $\widehat{\theta} = 18.5$. The nonparametric bootstrap treats the empirical distribution as the data-generating distribution:

```
For b = 1,...,B:
    Sample 6 observations with replacement from {12,15,18,19,21,44}
    Compute the median theta_b^* of the bootstrap sample
Use theta_1^*,...,theta_B^* to estimate bias and standard error
```

The procedure estimates how the median varies under repeated samples drawn from the empirical distribution.

(b) Let $\widehat{\theta}_b^*$ be the b-th bootstrap median and

$$\bar{\theta}^* = \frac{1}{B} \sum_{b=1}^{B} \widehat{\theta}_b^*.$$

The bootstrap bias estimate is

$$\widehat{\text{bias}}_B = \bar{\theta}^* - \widehat{\theta}.$$

The bootstrap standard error is

$$\widehat{\text{se}}_B = \left\{ \frac{1}{B-1} \sum_{b=1}^{B} (\widehat{\theta}_b^* - \bar{\theta}^*)^2 \right\}^{1/2}.$$

(c) The percentile interval takes the empirical bootstrap quantiles directly. Hence

$$CI_{0.95}^{perc} = [q_{0.025}, q_{0.975}] = [15, 32.5].$$

(d) With $n = 6$, there are $6^6 = 46656$ ordered bootstrap samples. If only the counts of how often each original observation appears matter, the number of distinct count vectors is

$$\binom{n+n-1}{n} = \binom{11}{6} = 462.$$

Thus $B = 10000$ can be computationally large relative to the number of distinct unordered resampling patterns. It is not wrong, but after many replications one mainly repeats patterns already seen.

Question 3.

(a) If $X_i \sim U[-1,1]$ iid, then

$$\theta = \mathbb{E}_\pi[e^X] = \int_{-1}^{1} e^x \frac{1}{2}\, dx.$$

The vanilla estimator is

$$\widehat{\theta}_N = \frac{1}{N}\sum_{i=1}^{N} e^{X_i}.$$

It is unbiased because

$$\mathbb{E}[\widehat{\theta}_N] = \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}[e^{X_i}] = \theta.$$

(b) Use the centred control variate $h(X) = X$, since $\mathbb{E}_\pi[X] = 0$. For a coefficient β,

$$\widehat{\theta}_{CV} = \frac{1}{N}\sum_{i=1}^{N}\{e^{X_i} - \beta X_i\}.$$

It remains unbiased because $\mathbb{E}[X] = 0$. The variance-minimising coefficient is

$$\beta^* = \frac{\mathrm{Cov}(e^X, X)}{\mathrm{Var}(X)}.$$

Here

$$\mathrm{Cov}(e^X, X) = \mathbb{E}[Xe^X] - \mathbb{E}[e^X]\mathbb{E}[X] = \mathbb{E}[Xe^X],$$

and

$$\mathbb{E}[Xe^X] = \frac{1}{2}\int_{-1}^{1} xe^x\, dx = \frac{1}{2}[(x-1)e^x]_{-1}^{1} = \frac{1}{e}.$$

Since $\mathrm{Var}(X) = 1/3$, $\beta^* = 3/e$.

(c) By symmetry,

$$\mathbb{E}[X] = \frac{1}{2}\int_{-1}^{1} x\, dx = 0.$$

Moreover,

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{1}{2}\int_{-1}^{1} x^2\, dx = \frac{1}{2}\cdot\frac{2}{3} = \frac{1}{3}.$$

(d) The function e^X is increasing and approximately linear over a short interval, so it is strongly positively correlated with X. Subtracting a multiple of the centred variable X removes part of the linear variation in e^X, leaving a less variable residual.

Complete Solution – Mock Exam 3

Question 1.

(a) For states $x_0 < \cdots < x_K$ with probabilities p_j, form cumulative probabilities

$$F_j = \sum_{k=0}^{j} p_k.$$

Draw $U \sim U(0,1)$ and return the first state x_j such that $U \leq F_j$. This works because the interval $(F_{j-1}, F_j]$ has length p_j.

(b) The cumulative probabilities are

$$F_0 = 0.1, \qquad F_1 = 0.3, \qquad F_2 = 0.7, \qquad F_3 = 1.$$

Since

$$0.3 < 0.63 \leq 0.7,$$

the inverse-transform rule returns $X = 2$.

(c) Pseudocode:

```
Draw U from Uniform(0,1)
s = 0
For j in {0,1,2,3}:
    s = s + p_j
    If U <= s:
        Return j
```

(d) The naive method checks states sequentially and therefore has cost $O(K)$ if there are K states. For a large distribution, this is slow if many samples are needed. Faster alternatives include binary search on the cumulative table, alias sampling, or specialised methods exploiting structure.

Question 2.

(a) The normalising constant is

$$Z = \int_0^1 x^2(1-x)^2\,dx = \int_0^1 (x^2 - 2x^3 + x^4)\,dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{5} = \frac{1}{30}.$$

Thus the normalised density is

$$\pi(x) = \frac{\gamma(x)}{Z} = 30x^2(1-x)^2\mathbf{1}_{[0,1]}(x).$$

This is Beta$(3,3)$, since $B(3,3) = 1/30$.

(b) With proposal $q(x) = 1$ on $[0,1]$, we need

$$\pi(x) \leq Mq(x) = M \quad \text{for all } x \in [0,1].$$

The maximum of $x^2(1-x)^2 = [x(1-x)]^2$ is reached at $x = 1/2$, where $x(1-x) = 1/4$. Hence

$$M = 30 \cdot \frac{1}{16} = \frac{15}{8}.$$

(c) In rejection sampling with a normalised target and proposal, the average acceptance probability is $1/M$. Therefore

$$p_{acc} = \frac{1}{M} = \frac{8}{15}.$$

Equivalently, the pointwise acceptance probability is

$$\frac{\pi(x)}{Mq(x)} = \frac{30x^2(1-x)^2}{15/8} = 16x^2(1-x)^2.$$

(d) Pseudocode:

```
Repeat:
    Draw X from Uniform(0,1)
    Draw U from Uniform(0,1)
    If U <= 16*X^2*(1-X)^2:
        Return X
```

The accepted X has density proportional to $q(x)\pi(x)/(Mq(x)) = \pi(x)$.

Question 3.

(a) If $U \sim U(0,1)$ and

$$X = \tan\{\pi(U - 1/2)\},$$

then X has the standard Cauchy distribution. This follows because the standard Cauchy cdf is

$$F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x,$$

whose inverse is $F^{-1}(u) = \tan\{\pi(u - 1/2)\}$.

(b) The line $y = x[x > 3]$ selects simulated Cauchy values larger than 3. Therefore

$$p_hat = \frac{\#\{i : X_i > 3\}}{N}$$

estimates the tail probability $\mathbb{P}(X > 3)$ for $X \sim \text{Cauchy}(0,1)$.

(c) Using the cdf,

$$\mathbb{P}(X > 3) = 1 - F(3) = 1 - \left(\frac{1}{2} + \frac{1}{\pi}\arctan 3\right) = \frac{1}{2} - \frac{1}{\pi}\arctan 3.$$

(d) Since the event $X > 3$ is a tail event, one can reduce variance by importance sampling with a proposal that samples more often in the right tail, or by simulating directly from the conditional tail distribution and correcting with the appropriate probability factor.

Complete Solution – Mock Exam 4

Question 1.

(a) The joint density factors as

$$\pi(x,y) = f_X(x) f_{Y|X=x}(y) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \cdot \frac{1}{\sqrt{2\pi}} e^{-(y-x)^2/2}.$$

Therefore

$$\pi(x,y) = \frac{1}{2\pi} \exp\left\{ -\frac{1}{2}x^2 - \frac{1}{2}(y-x)^2 \right\}.$$

(b) Use the hierarchical definition directly:

```
Draw X from N(0,1)
Draw epsilon from N(0,1), independently
Set Y = X + epsilon
Return (X,Y)
```

Then $Y \mid X = x = x + \epsilon \sim N(x, 1)$.

(c) Since $Y = X + \epsilon$, where X and ϵ are independent $N(0,1)$ variables,

$$\mathbb{E}[Y] = 0, \qquad \mathrm{Var}(Y) = \mathrm{Var}(X) + \mathrm{Var}(\epsilon) = 1 + 1 = 2.$$

Thus

$$Y \sim N(0, 2).$$

(d) Terms depending on x in the joint exponent are

$$-\frac{1}{2}\{x^2 + (y-x)^2\} = -\frac{1}{2}\{2x^2 - 2xy + y^2\}.$$

Complete the square:

$$2x^2 - 2xy + y^2 = 2\left(x - \frac{y}{2}\right)^2 + \frac{y^2}{2}.$$

Hence, as a function of x,

$$\pi(x \mid y) \propto \exp\left\{ -\left(x - \frac{y}{2}\right)^2 \right\}.$$

The density $N(m, v)$ has exponent $-(x-m)^2/(2v)$, so $1/(2v) = 1$ and $v = 1/2$. Therefore

$$X \mid Y = y \sim N\left(\frac{y}{2}, \frac{1}{2}\right).$$

Question 2.

(a) Let $X \sim \mathrm{Exp}(1)$, with density $e^{-x}\mathbf{1}_{x \geq 0}$. Then

$$I = \int_0^\infty e^{-x} \cos x \, dx = \mathbb{E}[\cos X].$$

Thus the test function is $\varphi(x) = \cos x$.

(b) With $X_1, \ldots, X_N \sim \mathrm{Exp}(1)$ iid,

$$\widehat{I}_N = \frac{1}{N} \sum_{i=1}^N \cos X_i.$$

Its variance is

$$\mathrm{Var}(\widehat{I}_N) = \frac{1}{N} \mathrm{Var}(\cos X) = \frac{1}{N}\{\mathbb{E}[\cos^2 X] - (\mathbb{E}[\cos X])^2\}.$$

(c) Use the identity

$$\int_0^\infty e^{-x} e^{ix} \, dx = \int_0^\infty e^{-(1-i)x} \, dx = \frac{1}{1-i}.$$

Taking real parts gives

$$I = \Re\left(\frac{1}{1-i}\right) = \Re\left(\frac{1+i}{2}\right) = \frac{1}{2}.$$

Also

$$\mathbb{E}[\cos^2 X] = \frac{1}{2} + \frac{1}{2}\mathbb{E}[\cos(2X)].$$

For $X \sim \text{Exp}(1)$, $\mathbb{E}[\cos(tX)] = 1/(1+t^2)$, hence $\mathbb{E}[\cos(2X)] = 1/5$ and

$$\mathbb{E}[\cos^2 X] = \frac{1}{2} + \frac{1}{10} = \frac{3}{5}.$$

So

$$\text{Var}(\widehat{I}_N) = \frac{1}{N}\left(\frac{3}{5} - \frac{1}{4}\right) = \frac{7}{20N}.$$

(d) Sampling directly from the exponential density incorporates the whole semi-infinite domain $[0, \infty)$ and the factor e^{-x} exactly. Truncating at a large upper bound replaces the original integral by an approximation and introduces truncation bias unless the tail error is explicitly bounded.

Question 3.

(a) Let π be the $N(0,1)$ density and q be the $N(\mu,1)$ density. If $X_i \sim q$, then

$$\widehat{p}_N = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{X_i > 4\}} w(X_i), \qquad w(x) = \frac{\pi(x)}{q(x)}.$$

(b) Since

$$\pi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}, \qquad q(x) = \frac{1}{\sqrt{2\pi}} e^{-(x-\mu)^2/2},$$

we get

$$w(x) = \exp\left\{-\frac{x^2}{2} + \frac{(x-\mu)^2}{2}\right\} = \exp\left(-\mu x + \frac{\mu^2}{2}\right).$$

(c) The event $Z > 4$ is rare under $N(0,1)$. Choosing $\mu > 0$ shifts the proposal to the right, so more simulated values fall in the important region $x > 4$. The weights then correct the bias introduced by sampling from the shifted distribution.

(d) If μ is too large, the proposal may oversample values far beyond 4 and assign unstable weights to the relevant region. Variance can increase because a few observations with large weights may dominate the estimator. Thus the proposal should shift mass toward the rare event without making the weight distribution too variable.

Complete Solution – Mock Exam 5

Question 1.

(a) The polar variables are $R = \sqrt{U_1}$ and $\Theta = 2\pi U_2$. For $0 \le r \le 1$,

$$\mathbb{P}(R \le r) = \mathbb{P}(\sqrt{U_1} \le r) = \mathbb{P}(U_1 \le r^2) = r^2,$$

so $f_R(r) = 2r$. Also $\Theta \sim U(0, 2\pi)$, independent of R. The joint density of (R, Θ) is

$$f_{R,\Theta}(r, \theta) = 2r \cdot \frac{1}{2\pi} = \frac{r}{\pi}.$$

The polar-to-Cartesian Jacobian is r, so

$$f_{X,Y}(x, y) = \frac{f_{R,\Theta}(r, \theta)}{r} = \frac{1}{\pi}$$

inside the unit disk. Hence (X, Y) is uniform over an area of size π.

(b) Uniformity on the disk means uniformity with respect to area, not radius. The area of the disk of radius r is πr^2, so the radial cdf must be r^2. Taking $R = U_1$ would give cdf r, putting too much probability near the centre relative to area.

(c) Pseudocode:

```
For i = 1,...,N:
    Draw U1, U2 independently from Uniform(0,1)
    R = sqrt(U1)
    Theta = 2*pi*U2
    X = R*cos(Theta)
    Y = R*sin(Theta)
    Store (X,Y)
```

(d) To sample uniformly on the disk of radius ρ centred at (a, b), first generate a uniform point (X, Y) on the unit disk and then scale and translate:

$$(X', Y') = (a + \rho X,\ b + \rho Y).$$

Equivalently, use

$$X' = a + \rho\sqrt{U_1}\cos(2\pi U_2), \qquad Y' = b + \rho\sqrt{U_1}\sin(2\pi U_2).$$

Question 2.

(a) The leave-one-out statistic $T_{(-i)}$ is obtained by removing observation i and recomputing T on the remaining $n - 1$ observations. The jackknife average is

$$\overline{T}_{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n} T_{(-i)}.$$

(b) The jackknife bias estimate is

$$\widehat{\text{bias}}_{jack} = (n-1)(\overline{T}_{(\cdot)} - T).$$

The jackknife standard error estimate is

$$\widehat{\text{se}}_{jack} = \left\{\frac{n-1}{n}\sum_{i=1}^{n}(T_{(-i)} - \overline{T}_{(\cdot)})^2\right\}^{1/2}.$$

(c) For data $2, 4, 5, 9$, the full-sample mean is $T = (2 + 4 + 5 + 9)/4 = 5$. The leave-one-out means are

$$T_{(-1)} = \frac{4+5+9}{3} = 6, \quad T_{(-2)} = \frac{2+5+9}{3} = \frac{16}{3},$$

$$T_{(-3)} = \frac{2+4+9}{3} = 5, \quad T_{(-4)} = \frac{2+4+5}{3} = \frac{11}{3}.$$

(d) Their average is

$$\overline{T}_{(\cdot)} = \frac{1}{4}\left(6 + \frac{16}{3} + 5 + \frac{11}{3}\right) = 5.$$

The jackknife variance estimate is

$$\widehat{\mathrm{Var}}_{jack}(T) = \frac{3}{4}\left[(6-5)^2 + \left(\frac{16}{3}-5\right)^2 + (5-5)^2 + \left(\frac{11}{3}-5\right)^2\right].$$

Thus

$$\widehat{\mathrm{Var}}_{jack}(T) = \frac{3}{4}\left[1 + \frac{1}{9} + 0 + \frac{16}{9}\right] = \frac{3}{4}\cdot\frac{26}{9} = \frac{13}{6}.$$

The jackknife standard error is $\sqrt{13/6}$.

Question 3.

(a) The unnormalised target is $\gamma(x) = e^{-x^2/2}$ and the proposal is $q_\lambda(x) = \frac{\lambda}{2}e^{-\lambda|x|}$. Therefore

$$\frac{\gamma(x)}{q_\lambda(x)} = \frac{2}{\lambda}\exp\left(-\frac{x^2}{2} + \lambda|x|\right).$$

(b) By symmetry it is enough to consider $x \geq 0$. Then the exponent is

$$g(x) = -\frac{x^2}{2} + \lambda x.$$

Differentiate:

$$g'(x) = -x + \lambda.$$

The critical point is $x = \lambda$, and $g''(x) = -1 < 0$, so it is the maximum on $x \geq 0$.

(c) Substituting $x = \lambda$ gives

$$M_\lambda = \max_x \frac{\gamma(x)}{q_\lambda(x)} = \frac{2}{\lambda}\exp\left(-\frac{\lambda^2}{2} + \lambda^2\right) = \frac{2}{\lambda}e^{\lambda^2/2}.$$

This is the smallest rejection constant for the unnormalised target.

(d) A reasonable choice balances tail coverage and acceptance rate. Minimising M_λ is equivalent to minimising

$$\log M_\lambda = \log 2 - \log\lambda + \frac{\lambda^2}{2}.$$

Differentiate:

$$\frac{d}{d\lambda}\log M_\lambda = -\frac{1}{\lambda} + \lambda.$$

Setting it equal to zero gives $\lambda^2 = 1$, hence $\lambda = 1$. This gives the best rejection constant within this family.

Complete Solution – Mock Exam 6

Question 1.

(a) The model is hierarchical. A composition algorithm is:

```
Draw P from Beta(alpha,beta)
Draw X from Binomial(m,P)
Return X
```

The second draw is conditional on the value of P generated in the first step.

(b) For $k = 0, 1, \ldots, m$,

$$\mathbb{P}(X = k) = \int_0^1 \mathbb{P}(X = k \mid P = p) f_P(p)\, dp.$$

Substituting the two densities,

$$\mathbb{P}(X = k) = \binom{m}{k} \int_0^1 p^k (1-p)^{m-k} \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha,\beta)}\, dp.$$

Equivalently,

$$\mathbb{P}(X = k) = \binom{m}{k} \frac{B(k+\alpha, m-k+\beta)}{B(\alpha,\beta)}.$$

(c) It is a discrete mixture because X is discrete conditionally on P, and its marginal probabilities are obtained by averaging the binomial probabilities over the continuous distribution of P.

(d) If $m = 1$, then $X \mid P = p \sim \text{Bernoulli}(p)$. Hence

$$\mathbb{P}(X = 1) = \mathbb{E}[\mathbb{P}(X = 1 \mid P)] = \mathbb{E}[P] = \frac{\alpha}{\alpha+\beta}.$$

Question 2.

(a) Since $U \sim U(0,1)$,

$$\theta = \mathbb{E}[\sqrt{U}] = \int_0^1 u^{1/2}\, du = \left[\frac{2}{3}u^{3/2}\right]_0^1 = \frac{2}{3}.$$

(b) With $U_1, \ldots, U_N \sim U(0,1)$ iid,

$$\widehat{\theta}_N = \frac{1}{N}\sum_{i=1}^{N} \sqrt{U_i}.$$

It is unbiased and has variance $\text{Var}(\sqrt{U})/N$.

(c) Divide $[0,1]$ into L strata

$$I_\ell = \left(\frac{\ell-1}{L}, \frac{\ell}{L}\right], \qquad \ell = 1, \ldots, L.$$

With one draw per stratum, draw $V_\ell \sim U(0,1)$ and set

$$U_\ell = \frac{\ell - 1 + V_\ell}{L}.$$

The stratified estimator is

$$\widehat{\theta}_{str} = \frac{1}{L}\sum_{\ell=1}^{L} \sqrt{U_\ell}.$$

(d) For a monotone function, points within the same stratum have less variation than points over the whole interval. Stratification forces the simulation to cover all parts of $[0,1]$ rather than leaving gaps, reducing sampling variability.

Question 3.

(a) The output line

```
np.mean(theta_star) - theta_hat # 0.35
```

gives the bootstrap estimate of bias. Therefore the estimated bias is 0.35.

(b) The standard error estimate is the sample standard deviation of the bootstrap statistics. The output gives

$$\widehat{\text{se}}_B = 1.92.$$

(c) The percentile interval uses the empirical 2.5% and 97.5% quantiles of the bootstrap medians:

$$[4.0, 11.0].$$

The median of the bootstrap distribution is 6.5, equal to the observed statistic $\theta_{hat} = 6.5$ in the output.

(d) The bootstrap distribution describes the sampling variability of the median under resampling from the empirical distribution. A positive bias estimate 0.35 means the bootstrap medians average slightly above the observed median. The interval $[4.0, 11.0]$ is fairly wide, indicating noticeable uncertainty in the median for the sample size used.

Complete Solution – Mock Exam 7

Question 1.

(a) For $-1 \le x \le 1$,

$$F(x) = \int_{-1}^{x} \frac{3}{8}(t+1)^2\,dt = \frac{3}{8}\left[\frac{(t+1)^3}{3}\right]_{-1}^{x} = \frac{(x+1)^3}{8}.$$

Also $F(x) = 0$ for $x < -1$ and $F(x) = 1$ for $x > 1$.

(b) Set $u = F(x)$:

$$u = \frac{(x+1)^3}{8} \quad \Longrightarrow \quad x + 1 = 2u^{1/3}.$$

Thus

$$F^{-1}(u) = 2u^{1/3} - 1.$$

(c) Pseudocode:

```
Draw U from Uniform(0,1)
X = 2*U^(1/3) - 1
Return X
```

(d) The density integrates to one because

$$\int_{-1}^{1} \frac{3}{8}(x+1)^2\,dx = \frac{3}{8}\left[\frac{(x+1)^3}{3}\right]_{-1}^{1} = \frac{3}{8}\cdot\frac{8}{3} = 1.$$

Question 2.

(a) If $U \sim U(0,1)$,

$$4I = 4\int_0^1 \frac{1}{1+x^2}\,dx = \mathbb{E}\left[\frac{4}{1+U^2}\right].$$

Since $I = \arctan(1) - \arctan(0) = \pi/4$, the quantity $4I$ equals π.

(b) The estimator is

$$\widehat{\pi}_N = \frac{1}{N}\sum_{i=1}^{N} \frac{4}{1+U_i^2}.$$

Let $G(U) = 4/(1+U^2)$. Then

$$\mathrm{Var}(\widehat{\pi}_N) = \frac{1}{N}\left(\mathbb{E}[G(U)^2] - \pi^2\right).$$

Now

$$\mathbb{E}[G(U)^2] = \int_0^1 \frac{16}{(1+x^2)^2}\,dx.$$

Using

$$\int \frac{dx}{(1+x^2)^2} = \frac{x}{2(1+x^2)} + \frac{1}{2}\arctan x,$$

we get

$$\mathbb{E}[G(U)^2] = 16\left(\frac{1}{4} + \frac{\pi}{8}\right) = 4 + 2\pi.$$

Therefore

$$\mathrm{Var}(\widehat{\pi}_N) = \frac{4 + 2\pi - \pi^2}{N}.$$

(c) The antithetic estimator with m pairs is

$$\widehat{\pi}_A = \frac{1}{m}\sum_{i=1}^{m} \frac{1}{2}\left\{\frac{4}{1+U_i^2} + \frac{4}{1+(1-U_i)^2}\right\}.$$

The function $4/(1+x^2)$ is decreasing on $[0,1]$, while the paired value uses $1 - U_i$, so the two evaluations tend to offset each other.

(d) From simulation output $G_i = 4/(1+U_i^2)$, estimate the standard error by

$$\widehat{\mathrm{se}}(\widehat{\pi}_N) = \frac{s_G}{\sqrt{N}}, \qquad s_G^2 = \frac{1}{N-1}\sum_{i=1}^{N}(G_i - \bar{G})^2.$$

For antithetic sampling, compute the sample standard deviation of the pair averages and divide by $\sqrt{m}$.

Question 3.

(a) The support condition is: whenever $\pi(x)|\varphi(x)| > 0$, we must have $q(x) > 0$. Equivalently, the proposal must cover all regions of the target that can contribute to the integral.

(b) Since $X_i \sim q$,

$$\mathbb{E}_q[w(X_i)\varphi(X_i)] = \int \frac{\pi(x)}{q(x)}\varphi(x)q(x)\,dx = \int \varphi(x)\pi(x)\,dx = \theta.$$

Averaging over i gives $\mathbb{E}[\widehat{\theta}_N] = \theta$.

(c) Taking $\varphi \equiv 1$ gives

$$\mathbb{E}_q[w(X)] = \int \frac{\pi(x)}{q(x)}q(x)\,dx = \int \pi(x)\,dx = 1,$$

because π is a normalised density.

(d) Standard IS requires normalised weights $w = \pi/q$ and is unbiased when π is known. SNIS uses unnormalised weights $\widetilde{w} = \gamma/q$ and estimates

$$\frac{\sum_i \widetilde{w}_i\varphi(X_i)}{\sum_i \widetilde{w}_i}.$$

SNIS can be used without knowing the normalising constant, but the ratio estimator is generally biased for finite N and consistent as $N \to \infty$.

Complete Solution – Mock Exam 8

Question 1.

(a) The conditional density is

$$f_c(x) = \frac{\phi(x)\mathbf{1}_{x>c}}{\mathbb{P}(X > c)} = \frac{\phi(x)\mathbf{1}_{x>c}}{1 - \Phi(c)},$$

where ϕ and Φ are the standard normal density and cdf.

(b) A naive rejection sampler draws $X \sim N(0,1)$ and accepts it if $X > c$. The acceptance probability is

$$p_{acc} = \mathbb{P}(X > c) = 1 - \Phi(c).$$

The accepted values have the desired conditional distribution.

(c) For large c, $1 - \Phi(c)$ is very small. Therefore the expected number of proposals is

$$\frac{1}{1 - \Phi(c)},$$

which becomes very large. Most normal draws are rejected because they fall below c.

(d) A better proposal should put mass on (c, ∞), for example a shifted exponential proposal on the tail:

$$Y = c + E, \qquad E \sim \text{Exp}(\lambda).$$

For rejection sampling, one needs a finite constant M such that

$$f_c(y) \leq Mq(y) \quad \text{for all } y > c.$$

Choosing a proposal with sufficiently heavy tail relative to the truncated normal tail is essential.

Question 2.

(a) The empirical cdf is

$$\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i \leq x\}}.$$

It is a step function that puts probability mass $1/n$ on each observation.

(b) The plug-in principle replaces the unknown distribution F by its empirical version $\widehat{F}_n$. Thus a functional $T(F)$ is estimated by $T(\widehat{F}_n)$.

(c) For a probability,

$$\widehat{\mathbb{P}}(X \leq t) = \widehat{F}_n(t) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i \leq t\}}.$$

For the mean,

$$\widehat{\mu} = \int x \, d\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

For the variance with denominator n,

$$\widehat{\sigma}^2 = \int (x - \widehat{\mu})^2 \, d\widehat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} (X_i - \widehat{\mu})^2.$$

(d) The empirical p-quantile is the smallest observed value x such that $\widehat{F}_n(x) \geq p$. In practice, sort the data and take the order statistic corresponding to the desired probability level, possibly with interpolation depending on convention.

Question 3.

(a) The algorithm is rejection sampling. It proposes from q and accepts a proposed value with probability $\gamma(X)/(Mq(X))$.

(b) The constant M must satisfy

$$\gamma(x) \leq Mq(x) \quad \text{for all } x$$

on the support of the target. Equivalently, $M \geq \sup_x \gamma(x)/q(x)$.

(c) The joint density of a proposed value x and acceptance is proportional to

$$q(x)\frac{\gamma(x)}{Mq(x)} = \frac{\gamma(x)}{M}.$$

After conditioning on acceptance, the normalising constant changes but the density becomes

$$\frac{\gamma(x)}{\int \gamma(t)\,dt},$$

which is the target density.

(d) The acceptance probability is

$$\int q(x)\frac{\gamma(x)}{Mq(x)}\,dx = \frac{1}{M}\int \gamma(x)\,dx.$$

If γ is already normalised, this is $1/M$. If γ is unnormalised with normalising constant Z, it is Z/M.

Complete Solution – Mock Exam 9

Question 1.

(a) Set $u = F(x) = 1/(1+e^{-x})$. Then

$$u(1+e^{-x}) = 1 \quad \Longrightarrow \quad ue^{-x} = 1-u \quad \Longrightarrow \quad e^{-x} = \frac{1-u}{u}.$$

Taking logarithms gives

$$x = \log \frac{u}{1-u}.$$

Thus $F^{-1}(u) = \log(u/(1-u))$.

(b) Pseudocode:

```
Draw U from Uniform(0,1)
X = log(U/(1-U))
Return X
```

(c) Differentiate $F(x) = (1+e^{-x})^{-1}$:

$$f(x) = F'(x) = \frac{e^{-x}}{(1+e^{-x})^2}.$$

Equivalently,

$$f(x) = \frac{e^{x}}{(1+e^{x})^2}.$$

(d) Inverse transform is convenient because the inverse cdf is explicit and uses only one uniform draw plus elementary functions. No rejection loop or numerical inversion is needed.

Question 2.

(a) A composition algorithm is:

```
Draw Lambda from Gamma(alpha,beta) with rate beta
Draw Y from Poisson(Lambda)
Return Y
```

(b) For $k = 0, 1, 2, \ldots$,

$$\mathbb{P}(Y=k) = \int_0^\infty \mathbb{P}(Y=k \mid \Lambda=\lambda) f_\Lambda(\lambda)\, d\lambda.$$

Substitute the Poisson and Gamma densities:

$$\mathbb{P}(Y=k) = \int_0^\infty \frac{e^{-\lambda}\lambda^k}{k!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}\, d\lambda.$$

(c) This is a Poisson–Gamma mixture. The marginal distribution is negative binomial, with parameters depending on the chosen negative-binomial convention.

(d) Conditional on Λ, Y has variance Λ. Marginally,

$$\mathbb{E}[Y] = \mathbb{E}[\Lambda], \qquad \mathrm{Var}(Y) = \mathbb{E}[\Lambda] + \mathrm{Var}(\Lambda).$$

The additional term $\mathrm{Var}(\Lambda)$ makes the variance larger than the mean, producing overdispersion relative to a Poisson with fixed parameter.

Question 3.

(a) The weights are $10, 1, 1, 1, 1, 1, 1, 1$. Their sum is

$$\sum_i w_i = 10 + 7 = 17.$$

Their squared sum is

$$\sum_i w_i^2 = 10^2 + 7 \cdot 1^2 = 107.$$

Therefore

$$\mathrm{ESS} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2} = \frac{17^2}{107} = \frac{289}{107} \approx 2.70.$$

(b) Although $N = 8$ samples were generated, the effective sample size is only about 2.7. One observation carries a much larger weight than the others, so the weighted average behaves as if it were based on far fewer equally weighted observations.

(c) Weight degeneracy increases variance because the estimator is dominated by a small number of high-weight samples. If those high-weight samples also have extreme values of $\varphi(X) = X^2$, the SNIS estimate can fluctuate substantially across repeated simulations.

(d) Improve the proposal by choosing q closer to the target in the regions that contribute most to $X^2\pi(x)$. In practice this means heavier tails or a proposal adapted to the target, so that large weights occur less frequently.

Complete Solution – Mock Exam 10

Question 1.

(a) The joint support is $0 < y < x < 1$. The marginal density of X is

$$f_X(x) = \int_0^x 2\,dy = 2x, \qquad 0 < x < 1.$$

(b) The conditional density is

$$f_{Y|X=x}(y) = \frac{f(x,y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x}, \qquad 0 < y < x.$$

Thus $Y \mid X = x \sim U(0,x)$.

(c) Since $f_X(x) = 2x$ on $[0,1]$, its cdf is $F_X(x) = x^2$, so $X = \sqrt{U}$. Then draw $V \sim U(0,1)$ and set $Y = xV$:

```
Draw U,V independently from Uniform(0,1)
X = sqrt(U)
Y = X*V
Return (X,Y)
```

(d) Check normalisation:

$$\int_0^1 \int_0^x 2\,dy\,dx = \int_0^1 2x\,dx = 1.$$

Therefore the proposed joint density integrates to one.

Question 2.

(a) Since $X \sim \text{Exp}(1)$,

$$\theta = \mathbb{P}(X > 2) = \int_2^\infty e^{-x}\,dx = e^{-2}.$$

The vanilla estimator is

$$\widehat{\theta}_N = \frac{1}{N}\sum_{i=1}^N \mathbf{1}_{\{X_i > 2\}}, \qquad X_i \sim \text{Exp}(1).$$

Its variance is

$$\text{Var}(\widehat{\theta}_N) = \frac{\theta(1-\theta)}{N} = \frac{e^{-2}(1-e^{-2})}{N}.$$

(b) With proposal $q_\lambda(x) = \lambda e^{-\lambda x}\mathbf{1}_{x \geq 0}$ and $X_i \sim q_\lambda$,

$$\widehat{\theta}_{IS} = \frac{1}{N}\sum_{i=1}^N \mathbf{1}_{\{X_i > 2\}} w_\lambda(X_i).$$

(c) The target density is $\pi(x) = e^{-x}\mathbf{1}_{x \geq 0}$, so

$$w_\lambda(x) = \frac{\pi(x)}{q_\lambda(x)} = \frac{e^{-x}}{\lambda e^{-\lambda x}} = \frac{1}{\lambda} e^{-(1-\lambda)x}, \qquad x \geq 0.$$

(d) For a right-tail probability, choose $\lambda < 1$. Then the proposal has mean $1/\lambda > 1$ and a heavier right tail than Exp(1), producing more samples with $X > 2$. If λ is too small, however, the proposal may put too much mass extremely far in the tail and weights may become unstable.

Question 3.

(a) For $q = N(m_q, v_q)$ and $p = N(m_p, v_p)$,

$$KL(q\|p) = \frac{1}{2}\left\{\log\frac{v_p}{v_q} + \frac{v_q + (m_q - m_p)^2}{v_p} - 1\right\}.$$

Here $m_q = ay + b$, $v_q = 2/3$, $m_p = y/2$, and $v_p = 1/2$.

(b) The terms involving a, b enter only through

$$(m_q - m_p)^2 = (ay + b - y/2)^2.$$

The variance terms are fixed because $v_q = 2/3$ and $v_p = 1/2$ are fixed.

(c) To minimise the KL divergence for every possible y, require the mean mismatch to be zero:

$$ay + b = \frac{y}{2} \quad \text{for all } y.$$

Matching coefficients gives

$$a = \frac{1}{2}, \qquad b = 0.$$

(d) If the variance of q were also free, the KL would be minimised by matching both mean and variance. Thus the best family member would have

$$m_q = m_p = \frac{y}{2}, \qquad v_q = v_p = \frac{1}{2}.$$

Complete Solution – Mock Exam 11

Question 1.

(a) The empirical cdf is

$$\widehat{F}_n(y) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{y_i \le y\}}.$$

Basic properties: it is nondecreasing, right-continuous, has jumps of size equal to empirical frequencies, tends to 0 as $y \to -\infty$, and tends to 1 as $y \to +\infty$.

(b) For data $3, 5, 5, 8, 11$, there are 5 observations. Hence

$$\widehat{F}_n(5) = \frac{\#\{3,5,5\}}{5} = \frac{3}{5},$$

$$\widehat{F}_n(7) = \frac{\#\{3,5,5\}}{5} = \frac{3}{5},$$

and

$$\widehat{F}_n(11) = \frac{\#\{3,5,5,8,11\}}{5} = 1.$$

(c) Plug-in for $\theta = \mathbb{P}(Y \le 6)$ gives

$$\widehat{\theta} = \widehat{F}_n(6) = \frac{3}{5}.$$

For the mean,

$$\widehat{\mu} = \frac{3+5+5+8+11}{5} = \frac{32}{5} = 6.4.$$

For variance with denominator n,

$$\widehat{\sigma}^2 = \frac{1}{5}\sum_{i=1}^{5}(y_i - 6.4)^2.$$

The squared deviations are $11.56, 1.96, 1.96, 2.56, 21.16$, whose sum is 39.2, so

$$\widehat{\sigma}^2 = \frac{39.2}{5} = 7.84.$$

(d) The empirical cdf estimates cumulative probabilities by putting point mass at observations. A histogram estimates the density by counting observations in bins. Narrow bins adapt to local features but have high variance; wide bins are smoother but introduce more bias. This is the bin-width bias–variance tradeoff.

Question 2.

(a) The recursion is $x_{k+1} = (5x_k + 1) \bmod 16$ with $x_0 = 3$. Hence

$$x_1 = (5 \cdot 3 + 1) \bmod 16 = 16 \bmod 16 = 0,$$

$$x_2 = (5 \cdot 0 + 1) \bmod 16 = 1,$$

$$x_3 = (5 \cdot 1 + 1) \bmod 16 = 6,$$

$$x_4 = (5 \cdot 6 + 1) \bmod 16 = 31 \bmod 16 = 15.$$

The corresponding uniforms are

$$u_1 = 0, \qquad u_2 = \frac{1}{16}, \qquad u_3 = \frac{6}{16}, \qquad u_4 = \frac{15}{16}.$$

(b) The period is the number of generated values before the deterministic sequence repeats. Since an LCG has finitely many states, it must eventually repeat; after repetition the whole sequence cycles.

(c) The code iterates the LCG

$$x_{k+1} = 65539 x_k \bmod 2^{31}$$

and stores $u_k = x_k / 2^{31}$. It then groups consecutive uniforms into triples

$$(u_i, u_{i+1}, u_{i+2}).$$

Low-quality LCGs can show lattice structure: points in two or three dimensions may lie on a small number of parallel lines or planes instead of filling the cube evenly.

(d) Visual inspection is useful for detecting obvious defects such as lattice planes or clustering. It is not sufficient because many defects appear only in higher dimensions, in long-run dependence, or under specific statistical tests.

Question 3.

(a) Let

$$\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\widehat{\theta}_b^*.$$

The bootstrap bias estimate is

$$\widehat{\text{bias}}_B = \bar{\theta}^* - \widehat{\theta}.$$

The bootstrap standard error is

$$\widehat{\text{se}}_B = \left\{\frac{1}{B-1}\sum_{b=1}^{B}(\widehat{\theta}_b^* - \bar{\theta}^*)^2\right\}^{1/2}.$$

(b) The percentile bootstrap confidence interval at nominal level 95% is

$$[q^*_{0.025}, q^*_{0.975}],$$

where q^*_p is the empirical p-quantile of the bootstrap replications.

(c) Given $\widehat{\theta} = 10$ and $\overline{\theta^*} = 10.4$,

$$\widehat{\text{bias}}_B = 10.4 - 10 = 0.4.$$

The standard error estimate is given as $s_{\theta^*} = 1.8$. The percentile interval is

$$[7.2, 13.9].$$

(d) Nonparametric bootstrap resamples the observed data with replacement and does not impose a parametric model. Parametric bootstrap first estimates a parametric distribution from the data and then simulates new samples from that fitted model.

Complete Solution – Mock Exam 12

Question 1.

(a) Since $X = \alpha U^{1/\beta}$, the support is $0 \le X \le \alpha$. For $0 \le x \le \alpha$,

$$F_X(x) = \mathbb{P}(\alpha U^{1/\beta} \le x) = \mathbb{P}\left(U \le \left(\frac{x}{\alpha}\right)^{\beta}\right) = \left(\frac{x}{\alpha}\right)^{\beta}.$$

Differentiating,

$$f_X(x) = \frac{\beta x^{\beta-1}}{\alpha^{\beta}}, \qquad 0 < x < \alpha.$$

(b) A density proportional to $x^{\beta-1}\mathbf{1}_{[0,\alpha]}(x)$ normalises to

$$f(x) = \frac{\beta x^{\beta-1}}{\alpha^{\beta}}.$$

The cdf is $(x/\alpha)^{\beta}$, so the inverse-transform algorithm is: draw $U \sim U(0,1)$ and return

$$X = \alpha U^{1/\beta}.$$

(c) With $\alpha = 2$ and $\beta = 3$,

$$\mathbb{P}(X \le 1) = F_X(1) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

(d) Inverse transform is exact when the inverse cdf is available and evaluated exactly. It may be inconvenient when the cdf has no closed form, when the inverse cdf has no closed form, or when numerical inversion must be repeated many times.

Question 2.

(a) A sum of α iid $\mathrm{Exp}(1)$ variables has Gamma distribution with shape α and rate 1:

$$S \sim \mathrm{Gamma}(\alpha, 1).$$

Similarly,

$$T \sim \mathrm{Gamma}(\beta, 1).$$

The variables S and T are independent.

(b) The joint density is

$$f_{S,T}(s,t) = \frac{s^{\alpha-1}e^{-s}}{\Gamma(\alpha)}\frac{t^{\beta-1}e^{-t}}{\Gamma(\beta)}, \qquad s,t > 0.$$

Use the transformation

$$r = \frac{s}{s+t}, \qquad v = s + t,$$

so that $s = rv$, $t = (1-r)v$, $0 < r < 1$, $v > 0$. The Jacobian is v. Hence

$$f_{R,V}(r,v) = \frac{(rv)^{\alpha-1}((1-r)v)^{\beta-1}e^{-v}}{\Gamma(\alpha)\Gamma(\beta)}v.$$

Collecting terms,

$$f_{R,V}(r,v) = \frac{r^{\alpha-1}(1-r)^{\beta-1}v^{\alpha+\beta-1}e^{-v}}{\Gamma(\alpha)\Gamma(\beta)}.$$

Integrating out v gives

$$f_R(r) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}r^{\alpha-1}(1-r)^{\beta-1},$$

so $R \sim \mathrm{Beta}(\alpha, \beta)$.

(c) Pseudocode:

```
Draw Y_1,...,Y_alpha iid Exp(1)
Draw Z_1,...,Z_beta iid Exp(1)
S = sum(Y_1,...,Y_alpha)
T = sum(Z_1,...,Z_beta)
Return R = S/(S+T)
```

(d) If $\alpha = \beta = 1$, the beta law is $\mathrm{Beta}(1,1) = U(0,1)$. Direct inverse transform simply returns one uniform random variable. The exponential-ratio method is correct but less efficient because it requires two exponential draws and a ratio.

Question 3.

(a) If $X \sim U(0,1)$, then

$$I = \int_0^1 x^2 e^{-x}\, dx = \mathbb{E}[X^2 e^{-X}].$$

The test function is $\varphi(x) = x^2 e^{-x}$.

(b) The vanilla estimator is

$$\widehat{I}_N = \frac{1}{N} \sum_{i=1}^{N} X_i^2 e^{-X_i}, \qquad X_i \sim U(0,1).$$

(c) Use the centred control variate $h(X) = X - 1/2$, since $\mathbb{E}[X - 1/2] = 0$. Then

$$\widehat{I}_{CV} = \frac{1}{N} \sum_{i=1}^{N} \{X_i^2 e^{-X_i} - \beta(X_i - 1/2)\}.$$

The optimal coefficient is

$$\beta^* = \frac{\mathrm{Cov}(X^2 e^{-X}, X - 1/2)}{\mathrm{Var}(X - 1/2)}.$$

Since $\mathrm{Var}(X - 1/2) = \mathrm{Var}(X) = 1/12$, this can also be written as

$$\beta^* = 12\ \mathrm{Cov}(X^2 e^{-X}, X - 1/2).$$

(d) The code first computes the integrand values $\phi = x^2 e^{-x}$. Then it computes the centred control $h = x - 0.5$. The line

```
beta = cov(phi,h)/var(h)
```

estimates β^* from the simulated sample. Finally,

```
est_cv = mean(phi - beta*h)
```

computes the sample average of the control-variate corrected observations.

Complete Solution – Mock Exam 13

Question 1.

(a) The beta density is

$$f(x) = \frac{x^{a-1}(1-x)^{b-1}}{B(a,b)}, \qquad 0 < x < 1.$$

For $a = b = 2$, $B(2,2) = 1/6$, so

$$f(x) = 6x(1-x).$$

Thus π is Beta$(2,2)$.

(b) With $q(x) = 1$ on $[0,1]$, need $\pi(x) \le M$. The function $x(1-x)$ is maximised at $x = 1/2$, with value $1/4$. Therefore

$$M = 6 \cdot \frac{1}{4} = \frac{3}{2}.$$

(c) The rejection acceptance rate is

$$p_{acc} = \frac{1}{M} = \frac{2}{3}.$$

The pointwise acceptance probability is

$$\frac{\pi(x)}{Mq(x)} = \frac{6x(1-x)}{3/2} = 4x(1-x).$$

(d) Pseudocode:

```
Repeat:
   Draw X from Uniform(0,1)
   Draw U from Uniform(0,1)
   If U <= 4*X*(1-X):
      Return X
```

Question 2.

(a) The unnormalised target is $\gamma(x) = e^{-x^2/2}$ and the proposal is $q(x) = \frac{1}{2}e^{-|x|}$. Rejection sampling accepts a proposal $X \sim q$ if

$$U \le \frac{\gamma(X)}{Mq(X)}.$$

(b) The ratio is

$$\frac{\gamma(x)}{q(x)} = 2\exp\left(-\frac{x^2}{2} + |x|\right).$$

For $x \ge 0$, maximise $-x^2/2 + x$, whose derivative is $-x + 1$, so the maximum is at $x = 1$. By symmetry, also $x = -1$ is a maximiser.

(c) At $|x| = 1$,

$$M = 2\exp\left(-\frac{1}{2} + 1\right) = 2e^{1/2}.$$

This is the smallest valid constant for the unnormalised normal kernel.

(d) A normal proposal for a Laplace target is problematic because the Laplace density has heavier tails than the normal. Then the ratio $\gamma_{Laplace}(x)/q_{Normal}(x)$ grows without bound as $|x| \to \infty$, so no finite M exists.

Question 3.

(a) The code estimates

$$\mathbb{P}(Z > 5), \qquad Z \sim N(0,1).$$

The event is represented by `ind = (x > 5)`.

(b) The target density is standard normal $\pi(x) = \phi(x)$ and the proposal density is $q(x) = \phi(x-3)$, i.e. $N(3,1)$.

(c) The importance weight is

$$w(x) = \frac{\phi(x)}{\phi(x-3)} = \exp\left(-\frac{x^2}{2} + \frac{(x-3)^2}{2}\right),$$

because the normalising constants cancel. Thus

$$\log w(x) = -\frac{1}{2}x^2 + \frac{1}{2}(x-3)^2,$$

which is exactly the code line for `logw`.

(d) Under $N(0,1)$, $x > 5$ is extremely rare, so most vanilla indicators would be zero. Sampling from $N(3,1)$ makes values near and above 5 much more frequent. The weights correct the change of measure, typically reducing variance for this right-tail probability.

Complete Solution – Mock Exam 14

Question 1.

(a) Integrate over the support $0 < y < x < 1$:

$$\int_0^1 \int_0^x 3x \, dy \, dx = \int_0^1 3x^2 \, dx = 1.$$

Therefore the density is already normalised. If the coefficient were a generic constant c, normalisation would require $c/3 = 1$, so $c = 3$.

(b) The marginal density of X is

$$f_X(x) = \int_0^x 3x \, dy = 3x^2, \qquad 0 < x < 1.$$

(c) The conditional density is

$$f_{Y|X=x}(y) = \frac{f(x,y)}{f_X(x)} = \frac{3x}{3x^2} = \frac{1}{x}, \qquad 0 < y < x.$$

Thus $Y \mid X = x \sim U(0, x)$.

(d) Since $f_X(x) = 3x^2$, the cdf is $F_X(x) = x^3$, so $X = U^{1/3}$. Then, conditional on $X = x$, draw $Y = xV$ with $V \sim U(0,1)$:

```
Draw U,V independently from Uniform(0,1)
X = U^(1/3)
Y = X*V
Return (X,Y)
```

Question 2.

(a) Draw a component indicator C with

$$\mathbb{P}(C = 1) = 0.3, \qquad \mathbb{P}(C = 2) = 0.7.$$

If $C = 1$, draw $X \sim N(-2, 1)$; if $C = 2$, draw $X \sim N(1, 4)$. Return X.

(b) By the law of total expectation,

$$\mathbb{E}[X] = 0.3(-2) + 0.7(1) = -0.6 + 0.7 = 0.1.$$

(c) By the law of total variance,

$$\mathrm{Var}(X) = \mathbb{E}[\mathrm{Var}(X \mid C)] + \mathrm{Var}(\mathbb{E}[X \mid C]).$$

The first term is

$$0.3 \cdot 1 + 0.7 \cdot 4 = 3.1.$$

The conditional means are -2 and 1, and the overall mean is 0.1, so the second term is

$$0.3(-2 - 0.1)^2 + 0.7(1 - 0.1)^2 = 0.3(4.41) + 0.7(0.81) = 1.89.$$

Thus

$$\mathrm{Var}(X) = 3.1 + 1.89 = 4.99.$$

(d) In a mixture, each observation is generated from one randomly selected component. A weighted average of two already generated observations, such as $0.3X_1 + 0.7X_2$, is a different random variable with a different variance and shape. It does not reproduce the mixture density.

Question 3.

(a) If $X_i \sim q$ are iid Cauchy draws, then

$$Z = \int \gamma(x) \, dx = \int \frac{\gamma(x)}{q(x)} q(x) \, dx = \mathbb{E}_q \left[\frac{\gamma(X)}{q(X)} \right].$$

The estimator is

$$\widehat{Z}_N = \frac{1}{N} \sum_{i=1}^{N} \frac{\gamma(X_i)}{q(X_i)}.$$

(b) Since $q(x) = 1/[\pi(1+x^2)]$,

$$w(x) = \frac{\gamma(x)}{q(x)} = \pi(1+x^2)e^{-x^2/2}.$$

(c) Yes. Taking expectation under q,

$$\mathbb{E}_q[\widehat{Z}_N] = \mathbb{E}_q[w(X)] = \int w(x)q(x)\,dx = \int \gamma(x)\,dx = Z.$$

Thus the estimator is unbiased.

(d) The Cauchy proposal has polynomial tails, while the Gaussian kernel has exponentially decaying tails. Therefore the proposal is heavier-tailed than the target. This matters because a heavier-tailed proposal protects against missing tail regions and helps keep the importance ratio stable in the tails.

Complete Solution – Mock Exam 15

Question 1.

(a) For $x < x_m$, $F(x) = 0$. For $x \geq x_m$,

$$F(x) = \int_{x_m}^{x} \alpha x_m^{\alpha} t^{-\alpha-1}\, dt = \left[-x_m^{\alpha} t^{-\alpha}\right]_{x_m}^{x} = 1 - \left(\frac{x_m}{x}\right)^{\alpha}.$$

(b) Set $u = 1 - (x_m/x)^{\alpha}$. Then

$$1 - u = \left(\frac{x_m}{x}\right)^{\alpha} \quad \Longrightarrow \quad x = x_m(1-u)^{-1/\alpha}.$$

Pseudocode:

```
Draw U from Uniform(0,1)
X = x_m*(1-U)^(-1/alpha)
Return X
```

Since $1 - U \sim U(0,1)$, one may equivalently use $X = x_m U^{-1/\alpha}$.

(c) For $K \geq x_m$,

$$\mathbb{P}(X > K) = 1 - F(K) = \left(\frac{x_m}{K}\right)^{\alpha}.$$

(d) Pareto distributions are heavy-tailed. The mean is finite only if $\alpha > 1$ and the variance is finite only if $\alpha > 2$. When α is small, a few very large simulated values can dominate the sample mean, creating high Monte Carlo variability.

Question 2.

(a) The data are $4, 7, 9, 10, 20$ and $n = 5$. The mean is

$$\bar{x} = \frac{4 + 7 + 9 + 10 + 20}{5} = 10.$$

The variance with denominator n is

$$\widehat{\theta} = \frac{1}{5}\sum_{i=1}^{5}(x_i - 10)^2 = \frac{1}{5}(36 + 9 + 1 + 0 + 100) = \frac{146}{5} = 29.2.$$

(b) The jackknife replication $\widehat{\theta}_{(-i)}$ is obtained by deleting x_i and recomputing the variance functional on the remaining 4 observations, using denominator 4. For example, deleting 4 gives data $7, 9, 10, 20$ and their denominator-4 variance.

(c) The five leave-one-out variances are

$$25.25, \quad 33.6875, \quad 36.1875, \quad 36.5, \quad 5.25.$$

Their average is

$$\overline{\theta}_{(\cdot)} = \frac{25.25 + 33.6875 + 36.1875 + 36.5 + 5.25}{5} = 27.375.$$

The jackknife bias estimate is

$$\widehat{\text{bias}}_{jack} = (n-1)(\overline{\theta}_{(\cdot)} - \widehat{\theta}) = 4(27.375 - 29.2) = -7.3.$$

(d) The sample variance is a nonlinear statistic of the data because it depends on the sample mean. The jackknife shows how the statistic changes when each observation is removed, giving information about finite-sample bias and sensitivity to influential observations such as the value 20.

Question 3.

(a) Since $X \sim \text{Exp}(1)$,

$$\theta = \mathbb{E}[e^{-X}] = \int_0^{\infty} e^{-x} e^{-x}\, dx = \int_0^{\infty} e^{-2x}\, dx = \frac{1}{2}.$$

(b) The vanilla estimator is

$$\widehat{\theta}_N = \frac{1}{N}\sum_{i=1}^{N} e^{-X_i}, \qquad X_i \sim \text{Exp}(1).$$

The variance is

$$\text{Var}(\widehat{\theta}_N) = \frac{1}{N}\,\text{Var}(e^{-X}).$$

Now

$$\mathbb{E}[e^{-2X}] = \int_0^\infty e^{-2x}e^{-x}\,dx = \int_0^\infty e^{-3x}\,dx = \frac{1}{3}.$$

Therefore

$$\text{Var}(e^{-X}) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}, \qquad \text{Var}(\widehat{\theta}_N) = \frac{1}{12N}.$$

(c) The exponential inverse cdf is $F^{-1}(u) = -\log(1-u)$. Let

$$X = F^{-1}(U) = -\log(1-U), \qquad X' = F^{-1}(1-U) = -\log U.$$

The antithetic estimator with m pairs is

$$\widehat{\theta}_A = \frac{1}{m}\sum_{i=1}^{m} \frac{e^{-X_i} + e^{-X_i'}}{2}.$$

(d) Antithetic variables are useful when the transformed integrand is monotone in U. Here $e^{-F^{-1}(u)} = e^{\log(1-u)} = 1-u$, which is decreasing in u. Pairing U with $1-U$ offsets high and low values, producing negative dependence and reducing variance.

Complete Solution – Mock Exam 16

Question 1.

(a) For a Poisson random variable, compute cumulative probabilities

$$F(k) = \sum_{j=0}^{k} e^{-\lambda} \frac{\lambda^j}{j!}.$$

Draw $U \sim U(0,1)$ and return the smallest integer k such that $U \leq F(k)$.

(b) For $\lambda = 2$,

$$p_0 = e^{-2}, \quad p_1 = 2e^{-2}, \quad p_2 = \frac{2^2}{2!}e^{-2} = 2e^{-2}, \quad p_3 = \frac{2^3}{3!}e^{-2} = \frac{4}{3}e^{-2}.$$

Thus

$$F(0) = e^{-2} \approx 0.1353,$$
$$F(1) = 3e^{-2} \approx 0.4060,$$
$$F(2) = 5e^{-2} \approx 0.6767,$$
$$F(3) = \frac{19}{3}e^{-2} \approx 0.8571.$$

(c) Since

$$F(2) \approx 0.6767 < 0.80 \leq F(3) \approx 0.8571,$$

the sampled value is $X = 3$.

(d) The code starts with $p_0 = e^{-\lambda}$ and $F = p_0$. The while loop increases k until the cumulative probability F exceeds the uniform u. The recursion

$$p_k = p_{k-1}\frac{\lambda}{k}$$

updates Poisson probabilities without recomputing factorials.

Question 2.

(a)

$$I = \int_0^1 x^{1/2}\,dx = \left[\frac{2}{3}x^{3/2}\right]_0^1 = \frac{2}{3}.$$

(b) With $U_i \sim U(0,1)$ iid,

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^{N} \sqrt{U_i}.$$

(c) With two strata of equal length, draw $V_1, V_2 \sim U(0,1)$ independently and set

$$X_1 = \frac{V_1}{2} \in [0, 1/2], \qquad X_2 = \frac{1}{2} + \frac{V_2}{2} \in (1/2, 1].$$

The stratified estimator is

$$\widehat{I}_{str} = \frac{1}{2}\sqrt{X_1} + \frac{1}{2}\sqrt{X_2}.$$

Each stratum receives weight equal to its length, $1/2$.

(d) The function $\sqrt{x}$ is monotone increasing and smooth on most of the interval. Stratification forces one draw from the lower half and one from the upper half, reducing the chance that the sample overrepresents one part of the interval.

Question 3.

(a) Draw $Z_i \sim N(0,1)$ and compute

$$S_T^{(i)} = S_0 \exp\left((r - \sigma^2/2)T + \sigma\sqrt{T}Z_i\right).$$

The vanilla discounted call estimator is

$$\widehat{C}_N = e^{-rT}\frac{1}{N}\sum_{i=1}^{N}(S_T^{(i)} - K)^+.$$

(b) For each Z_i, also use $-Z_i$. Define

$$S_T^+(Z_i) = S_0 e^{(r-\sigma^2/2)T+\sigma\sqrt{T}Z_i},$$
$$S_T^-(Z_i) = S_0 e^{(r-\sigma^2/2)T-\sigma\sqrt{T}Z_i}.$$

The antithetic estimator is

$$\widehat{C}_A = e^{-rT}\frac{1}{m}\sum_{i=1}^{m}\frac{(S_T^+(Z_i)-K)^+ + (S_T^-(Z_i)-K)^+}{2}.$$

(c) Use S_T as a control variate. Under the risk-neutral dynamics,

$$\mathbb{E}[S_T] = S_0 e^{rT}.$$

A control-variate estimator can be written as

$$e^{-rT}\frac{1}{N}\sum_{i=1}^{N}\left[(S_T^{(i)}-K)^+ - \beta(S_T^{(i)} - S_0 e^{rT})\right],$$

with β estimated or chosen optimally.

(d) For out-of-the-money options, many simulated payoffs are zero and only relatively rare high values contribute to the price. This creates high variance. Variance reduction improves accuracy without requiring an extremely large number of simulations.

Complete Solution – Mock Exam 17

Question 1.

(a) The normalising constant is

$$Z = \int_0^2 e^{-x}\,dx = 1 - e^{-2}.$$

(b) For $0 < x < 2$,

$$F(x) = \int_0^x \frac{e^{-t}}{1-e^{-2}}\,dt = \frac{1-e^{-x}}{1-e^{-2}}.$$

Set $u = F(x)$:

$$u(1-e^{-2}) = 1 - e^{-x} \quad \Longrightarrow \quad e^{-x} = 1 - u(1-e^{-2}).$$

Therefore

$$F^{-1}(u) = -\log\{1 - u(1-e^{-2})\}.$$

(c) Pseudocode:

```
Draw U from Uniform(0,1)
X = -log(1 - U*(1-exp(-2)))
Return X
```

(d) Draw $Y \sim \text{Exp}(1)$ and accept if $0 < Y < 2$. Since the proposal density is e^{-y} on $y > 0$, accepted draws have density proportional to e^{-y} on $(0,2)$, which is the target. The acceptance probability is

$$\mathbb{P}(0 < Y < 2) = 1 - e^{-2} = Z.$$

The expected number of proposals is $1/(1-e^{-2})$.

Question 2.

(a) The data are $8, 9, 10, 13, 14, 60$. The value 60 is an outlier relative to the rest. The mean is strongly affected by this outlier, while the median depends only on the central order statistics, so it is more robust.

(b) Nonparametric bootstrap for the median:

```
For b = 1,...,B:
    Resample 6 observations with replacement from the data
    Compute the median med_b^*
Use the empirical distribution of med_b^* to estimate bias, se, and intervals
```

(c) The percentile interval is read from the quantiles:

$$[8.5, 37.0].$$

The bootstrap bias estimate is

$$\overline{med^*} - \widehat{med} = 3.1.$$

The bootstrap standard error is

$$10.4.$$

(d) With $n = 6$, there are $6^6 = 46656$ ordered bootstrap samples and only

$$\binom{6+6-1}{6} = \binom{11}{6} = 462$$

distinct unordered count patterns. Therefore $B = 4000$ is valid, but not all replications necessarily provide new patterns; many repeats can occur.

Question 3.

(a) The SNIS estimator is

$$\widehat{\theta}_{SNIS} = \frac{\sum_{i=1}^N \widetilde{w}_i \varphi(X_i)}{\sum_{i=1}^N \widetilde{w}_i}, \qquad \widetilde{w}_i = \frac{\gamma(X_i)}{q(X_i)}.$$

(b) If $\gamma(x) = Z\pi(x)$, then $\widetilde{w}_i = Z\pi(X_i)/q(X_i) = Zw_i$. In the SNIS ratio,

$$\frac{\sum_i Zw_i\varphi(X_i)}{\sum_i Zw_i},$$

the unknown constant Z cancels. Therefore SNIS can be used even when Z is unknown.

(c) SNIS is generally not unbiased for finite N because it is a ratio of two random sums:

$$\mathbb{E}\left[\frac{A_N}{B_N}\right] \neq \frac{\mathbb{E}[A_N]}{\mathbb{E}[B_N]}$$

in general. However, under standard regularity conditions it is consistent as $N \to \infty$.

(d) The effective sample size is commonly estimated by

$$\mathrm{ESS} = \frac{(\sum_i \widetilde{w}_i)^2}{\sum_i \widetilde{w}_i^2}.$$

It ranges from 1 to N. A value near N indicates balanced weights; a value near 1 indicates that one or very few samples dominate.

Complete Solution – Mock Exam 18

Question 1.

(a) Let $U, V \sim U(0,1)$ independently. Then

$$I = \int_0^1 \int_0^1 e^{xy}\, dx\, dy = \mathbb{E}[e^{UV}].$$

(b) Draw (U_i, V_i) iid with independent uniforms and define

$$\widehat{I}_N = \frac{1}{N} \sum_{i=1}^{N} e^{U_i V_i}.$$

It is unbiased because

$$\mathbb{E}[\widehat{I}_N] = \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}[e^{U_i V_i}] = \mathbb{E}[e^{UV}] = I.$$

(c) Let $G_i = e^{U_i V_i}$ and $\bar{G} = \widehat{I}_N$. The sample variance is

$$s_G^2 = \frac{1}{N-1} \sum_{i=1}^{N} (G_i - \bar{G})^2.$$

The estimated Monte Carlo standard error is

$$\widehat{\mathrm{se}}(\widehat{I}_N) = \frac{s_G}{\sqrt{N}}.$$

(d) Grid quadrature suffers from the curse of dimensionality: if m grid points are used per dimension, a d-dimensional grid uses m^d points. Monte Carlo error is typically $O(N^{-1/2})$ under finite variance, independent of dimension in the rate, although the variance constant may still depend on dimension.

Question 2.

(a) Since $\mathbb{E}_\pi[h(X)] = 0$, the control-variate estimator is

$$\widehat{\theta}_\beta = \frac{1}{N} \sum_{i=1}^{N} \{\varphi(X_i) - \beta h(X_i)\}, \qquad X_i \sim \pi.$$

It is unbiased for every fixed β.

(b) The variance of one corrected observation is

$$\mathrm{Var}(\varphi(X) - \beta h(X)) = \mathrm{Var}(\varphi) - 2\beta \,\mathrm{Cov}(\varphi, h) + \beta^2 \,\mathrm{Var}(h).$$

Differentiate with respect to β and set equal to zero:

$$-2\,\mathrm{Cov}(\varphi, h) + 2\beta\,\mathrm{Var}(h) = 0.$$

Therefore

$$\beta^* = \frac{\mathrm{Cov}(\varphi, h)}{\mathrm{Var}(h)}.$$

(c) Substituting β^* gives

$$\mathrm{Var}(\varphi - \beta^* h) = \mathrm{Var}(\varphi) - \frac{\mathrm{Cov}(\varphi, h)^2}{\mathrm{Var}(h)}.$$

In terms of the correlation ρ,

$$\mathrm{Var}(\varphi - \beta^* h) = \mathrm{Var}(\varphi)(1 - \rho^2).$$

The variance of the Monte Carlo average is this quantity divided by N.

(d) If h is weakly correlated with φ, then ρ^2 is close to zero, so the reduction factor $1 - \rho^2$ is close to one. In that case the control variate gives little improvement and may even be unhelpful if the coefficient is estimated noisily.

Question 3.

(a) For estimating $\theta = \int \varphi(x)\pi(x)\,dx$ with $\varphi \geq 0$, the zero-variance or ideal proposal is

$$q^*(x) = \frac{\varphi(x)\pi(x)}{\int \varphi(t)\pi(t)\,dt} = \frac{\varphi(x)\pi(x)}{\theta}.$$

Under this proposal, the importance weight times φ is constant and the variance is zero.

(b) The ideal proposal is usually not usable because it contains the unknown target quantity θ in its normalising constant. It may also be difficult to sample from q^* even if its unnormalised form is known.

(c) The forward KL criterion is

$$KL(\pi \| q_\phi) = \int \pi(x) \log \frac{\pi(x)}{q_\phi(x)}\,dx.$$

The reverse KL criterion is

$$KL(q_\phi \| \pi) = \int q_\phi(x) \log \frac{q_\phi(x)}{\pi(x)}\,dx.$$

(d) Forward KL tends to be mass-covering because it heavily penalises setting $q_\phi(x)$ small where $\pi(x)$ is large. Reverse KL tends to be mode-seeking because it penalises placing mass where $\pi(x)$ is small and may prefer focusing on one mode of a multimodal target.

Complete Solution – Mock Exam 19

Question 1.

(a) Let $Y = X^2$. For $y > 0$, the preimages are $x = \sqrt{y}$ and $x = -\sqrt{y}$. Using the transformation formula,

$$f_Y(y) = f_X(\sqrt{y}) \left| \frac{d}{dy}\sqrt{y} \right| + f_X(-\sqrt{y}) \left| \frac{d}{dy}(-\sqrt{y}) \right|.$$

Since $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$,

$$f_Y(y) = 2 \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi}} y^{-1/2} e^{-y/2}, \qquad y > 0.$$

(b) This is the χ^2_1 density. Equivalently, $Y \sim \text{Gamma}(1/2, 1/2)$ under the rate parametrisation.

(c) If $Z_1, \ldots, Z_n$ are iid $N(0,1)$, then

$$\sum_{i=1}^{n} Z_i^2 \sim \chi^2_n.$$

Thus a chi-square random variable with n degrees of freedom can be simulated by summing squares of n independent standard normals.

(d) Pseudocode:

```
S = 0
For i = 1,...,n:
    Draw Z_i from N(0,1)
    S = S + Z_i^2
Return S
```

Question 2.

(a) Use composition:

```
Draw Lambda from Gamma(alpha,beta) with rate beta
Draw X from Poisson(Lambda)
Return X
```

(b) For $k = 0, 1, 2, \ldots$,

$$\mathbb{P}(X = k) = \int_0^\infty \frac{e^{-\lambda}\lambda^k}{k!} \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}\, d\lambda.$$

Combining powers and exponentials,

$$\mathbb{P}(X = k) = \frac{\beta^\alpha}{k!\Gamma(\alpha)} \int_0^\infty \lambda^{k+\alpha-1} e^{-(\beta+1)\lambda}\, d\lambda.$$

This evaluates to a negative-binomial probability:

$$\mathbb{P}(X = k) = \frac{\Gamma(k+\alpha)}{k!\Gamma(\alpha)} \left(\frac{\beta}{\beta+1}\right)^\alpha \left(\frac{1}{\beta+1}\right)^k.$$

(c) By iterated expectation,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}(X \mid \Lambda)] = \mathbb{E}[\Lambda] = \frac{\alpha}{\beta}.$$

(d) By the law of total variance,

$$\text{Var}(X) = \mathbb{E}[\text{Var}(X \mid \Lambda)] + \text{Var}(\mathbb{E}[X \mid \Lambda]).$$

For a Poisson variable, conditional mean and variance are both Λ, so

$$\text{Var}(X) = \mathbb{E}[\Lambda] + \text{Var}(\Lambda) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

Question 3.

(a) The target density is $\pi(x) = e^{-x}\mathbf{1}_{x\geq 0}$. With $X_i \sim q_\lambda$,

$$\widehat{p}_{IS} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_{\{X_i > K\}} w_\lambda(X_i).$$

(b) The weight is

$$w_\lambda(x) = \frac{e^{-x}}{\lambda e^{-\lambda x}} = \frac{1}{\lambda} e^{-(1-\lambda)x}, \qquad x \geq 0.$$

(c) For a right-tail probability, choose $\lambda < 1$. Then q_λ has mean $1/\lambda > 1$ and a heavier tail than the original exponential, so tail exceedances $X > K$ occur more frequently.

(d) If λ is too small, the proposal becomes too heavy-tailed and produces many extremely large values. For such values the weight

$$w_\lambda(x) = \lambda^{-1} e^{-(1-\lambda)x}$$

can be very small, so many samples contribute almost nothing. The estimator may again become inefficient because simulation effort is wasted far beyond the important region.

Complete Solution – Mock Exam 20

Question 1.

(a) For $0 \le x \le 1$,

$$F_X(x) = \mathbb{P}(\sqrt{U} \le x) = \mathbb{P}(U \le x^2) = x^2.$$

Differentiating,

$$f_X(x) = 2x\mathbf{1}_{[0,1]}(x).$$

This is Beta$(2,1)$ because $B(2,1) = 1/2$ and the beta density is $2x$ on $[0,1]$.

(b) The sum of two independent Exp(1) variables is Gamma with shape 2 and rate 1. Directly, the convolution is

$$f_T(t) = \int_0^t e^{-y}e^{-(t-y)}\,dy = \int_0^t e^{-t}\,dy = te^{-t}, \qquad t \ge 0.$$

(c) Let $V = T + Y_3$ and $Z = T/(T + Y_3)$. The inverse transformation is

$$T = zv, \qquad Y_3 = (1-z)v,$$

with $0 < z < 1$, $v > 0$. The Jacobian is v. Since $f_T(t) = te^{-t}$ and $f_{Y_3}(y) = e^{-y}$,

$$f_{T,Y_3}(t,y) = te^{-t}e^{-y} = te^{-(t+y)}.$$

Substituting $t = zv$, $y = (1-z)v$,

$$f_{Z,V}(z,v) = (zv)e^{-v} \cdot v = zv^2e^{-v}.$$

Integrating out v,

$$f_Z(z) = z\int_0^\infty v^2e^{-v}\,dv = z\Gamma(3) = 2z, \qquad 0 < z < 1.$$

Thus $Z \sim$ Beta$(2,1)$.

(d) The method $X = \sqrt{U}$ is computationally preferable for Beta$(2,1)$ because it uses one uniform draw and one square root. The gamma-ratio method is more general but requires three exponential draws, sums, and a ratio, so it is more expensive for this special case.

Question 2.

(a) Let $X \sim U(0,1)$ and define

$$\varphi(X) = 4\sqrt{1-X^2}.$$

Then

$$I = 4\int_0^1 \sqrt{1-x^2}\,dx = \mathbb{E}[4\sqrt{1-X^2}].$$

The Monte Carlo estimator is

$$\widehat{I}_N = \frac{1}{N}\sum_{i=1}^N 4\sqrt{1-X_i^2}, \qquad X_i \sim U(0,1).$$

(b) Since $I = \pi$,

$$\mathrm{Var}(\widehat{I}_N) = \frac{1}{N}\left(\mathbb{E}[16(1-X^2)] - \pi^2\right).$$

Now

$$\mathbb{E}[16(1-X^2)] = 16\left(1 - \mathbb{E}[X^2]\right) = 16\left(1 - \frac{1}{3}\right) = \frac{32}{3}.$$

Therefore

$$\mathrm{Var}(\widehat{I}_N) = \frac{1}{N}\left(\frac{32}{3} - \pi^2\right).$$

(c) Use antithetic uniforms U and $1-U$:

$$\widehat{I}_A = \frac{1}{m}\sum_{i=1}^m \frac{4\sqrt{1-U_i^2} + 4\sqrt{1-(1-U_i)^2}}{2}.$$

This pairs points from opposite sides of the interval and can reduce variance because the integrand is monotone decreasing on $[0,1]$.

(d) Let the five estimates be $\theta_1, \ldots, \theta_5$. The jackknife leave-one-out averages are

$$\theta_{(-i)} = \frac{1}{4}\sum_{j \neq i} \theta_j, \qquad i = 1, \ldots, 5.$$

Then compute

$$\bar{\theta}_{(\cdot)} = \frac{1}{5}\sum_{i=1}^{5} \theta_{(-i)}$$

and estimate the variance of the average by

$$\widehat{\mathrm{Var}}_{jack} = \frac{5-1}{5}\sum_{i=1}^{5} (\theta_{(-i)} - \bar{\theta}_{(\cdot)})^2.$$

For these numbers, the sample mean is

$$\bar{\theta} = \frac{3.05 + 3.18 + 3.09 + 3.22 + 3.11}{5} = 3.13,$$

and the ordinary sample variance of the five estimates is

$$s^2 = \frac{(-0.08)^2 + (0.05)^2 + (-0.04)^2 + (0.09)^2 + (-0.02)^2}{4} = 0.00475.$$

Thus, for the mean,

$$\widehat{\mathrm{Var}}_{jack} = \frac{s^2}{5} = 0.00095.$$

Question 3.

(a) If $X_i \sim q$ and the support condition holds, the standard IS estimator is

$$\widehat{\theta}_{IS} = \frac{1}{N}\sum_{i=1}^{N} \varphi(X_i) w(X_i), \qquad w(x) = \frac{\pi(x)}{q(x)}.$$

(b) Since π is normalised,

$$\mathbb{E}_q[\widehat{\theta}_{IS}] = \frac{1}{N}\sum_{i=1}^{N} \int \varphi(x)\frac{\pi(x)}{q(x)} q(x)\, dx = \int \varphi(x)\pi(x)\, dx.$$

Therefore the standard IS estimator is unbiased.

(c) If only $\gamma(x) = Z\pi(x)$ is known, use unnormalised weights

$$\widetilde{w}(x) = \frac{\gamma(x)}{q(x)}.$$

The SNIS estimator is

$$\widehat{\theta}_{SNIS} = \frac{\sum_{i=1}^{N} \widetilde{w}(X_i)\varphi(X_i)}{\sum_{i=1}^{N} \widetilde{w}(X_i)}.$$

The unknown Z cancels from numerator and denominator.

(d) Standard IS needs the normalised target density π and is unbiased under the support condition. SNIS only needs an unnormalised density γ, but is generally biased for finite N because it is a ratio estimator. Both methods can suffer from weight degeneracy. ESS is especially useful with SNIS to diagnose how concentrated the normalised weights are.